Query-Conditioned Budget-Aware Evidence Allocation for Retrieval-Augmented Generation

Doddi Guna Venkat (251140009) **Gagan Kumar Soni (252040606)**

CS787: Generative Artificial Intelligence · Indian Institute of Technology Kanpur

GitHub: github.com/Guna-Venkat/QC-BEA-for-RAG

Abstract

Retrieval-augmented generation (RAG) must often select a small evidence context from a larger retrieved pool. We investigate how to allocate limited evidence tokens across sources, focusing on *which evidence to retain* and *at what granularity*. Starting from SARA's explicit-text versus latent-evidence trade-off, we find that retrospective query-specific context-size choices offer substantial headroom, but learned controllers frequently under-allocate. Controlled experiments on HotpotQA, QASPER, and SciFact then separate the effects of retrieval, reader behavior, passage identity, and sentence-level granularity. We propose QBGF (Query-Conditioned Budgeted Granularity Fusion), which uses a multiple-choice knapsack program to select an extractive representation for each passage under a fixed token budget. Although query conditioning consistently improves its own ablation, QBGF does not reliably outperform strong sentence selection; a pre-specified rule consequently retains the sentence baseline for reserved confirmation. Leave-one-out interventions reveal that semantic relevance often poorly predicts a unit's effect on the reader. We therefore study BEA-V1, an incremental allocator with a marginal-value stopping rule, and BEA-V2, which weights candidate gains using a classifier trained on historical leave-one-out helpfulness labels. BEA-V1 can stop prematurely, while exploratory BEA-V2 comparisons achieve the highest HotpotQA Answer F1 at both 256 and 1024 evidence tokens (0.4352 and 0.4234, versus 0.3661 and 0.3893 for sentence selection). Results on the scientific datasets remain mixed, and the BEA-V2 operating point lacks independent confirmation. The findings distinguish exact optimization of a relevance surrogate from reliable estimation of downstream evidence utility.

1 Introduction

Retrieval-augmented generation improves language models by conditioning generation on external evidence [1]. In practice, retrieved material can exceed the evidence budget we want the reader to process; longer prompts cost more and can make useful information harder to exploit [4]. Context-compression methods such as RECOMP therefore motivate selecting a smaller, task-relevant subset before generation [3].

Core question: Given a fixed evidence-token budget, which parts of the retrieved material should the reader see, and at what granularity?

We study three coupled decisions. **Quantity** asks how much context to allocate, **identity** asks which retrieved sources to retain, and **granularity** asks whether a source should appear as a whole passage, selected sentences, or a finer extractive span. We begin from SARA [2], where some evidence remains readable text and the rest is compressed into latent vectors, motivating the hypothesis that the amount of explicit text should vary by query. Figure 1 summarizes how successive experiments refine that hypothesis into a budgeted evidence-allocation problem.

Contributions and evidence in brief. Our contributions concern the formulation, controlled comparisons, and empirical diagnosis of budgeted evidence selection:

1. **Separate quantity, identity, and representation.** Starting from static SARA $k = 5$, we test adaptive context size, passage selection, and evidence granularity as distinct decisions. Retrospective context-size oracles expose headroom, but learned controllers under-allocate and do not reliably recover it; oracle gains are not deployable gains.
2. **QBGF: query-conditioned representation allocation.** We define exclusive choices to drop a passage, select 1, 2, 4, or 8 sentences, or keep it in full. Query-conditioned relevance mass, exact source serialization, and a token-budget DP define the concrete surrogate. Query conditioning improves the method relative to its no-query ablation, but the FULL option does not establish consistent superiority over sentence selection.
3. **Reader-utility diagnosis.** Matched leave-one-out interventions separate semantic relevance from the observed effect of a selected unit on the reader. Helpful, neutral, and harmful evidence reveal why exact optimization of a relevance surrogate can still optimize the wrong target. These are context-dependent interventions, not universal causal necessity labels.
4. **BEA-V1 and BEA-V2.** We implement incremental source-representation selection with context-dependent marginal transitions and stopping; V2 weights positive semantic gain by learned LOO helpfulness. V1 exposes harmful context starvation. V2 improves matched exploratory HotpotQA F1 from 0.3661 to 0.4352 at 256 tokens and from 0.3893 to 0.4234 at 1024, while QASPER/SciFact results remain mixed and corrected comparisons do not establish a new confirmed winner.

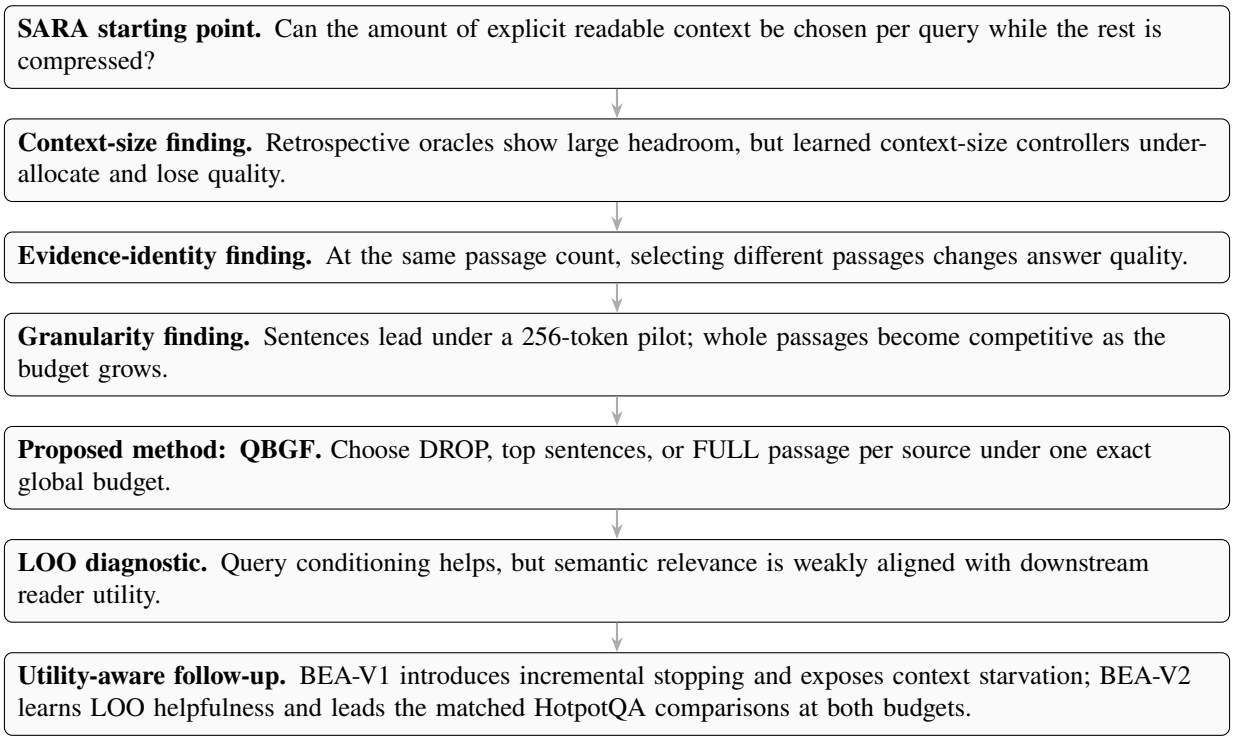


Figure 1: Research path from the SARA starting point to utility-aware evidence allocation. The stages motivate the progression from context quantity to evidence identity, representation, and downstream utility.

5. **A locked decision with informative negative results.** Retrieval/reader controls, grouped predictor folds, a pre-specified validation promotion rule, and reserved confirmation keep development findings distinct from generalization claims. Generative extraction, more elaborate selectors, and universal mixed-granularity promotion fail to deliver consistent gains; the simpler sentence policy remains the frozen final system.

2 Literature Survey and Positioning

2.1 Context compression: RECOMP and the SARA starting point

RECOMP [3] studies extractive sentence compression and abstractive summaries before reader generation, including omission when retrieved text contributes little. It motivates our comparison of suppression, extraction, and query-conditioned retention. Our fixed sentence baseline is not a reproduction of RECOMP's trained compressor, and we report no matched RECOMP quality score.

SARA [2] combines readable text with compressed semantic representations. It is the parent architecture for our early experiments: with $n = 10$, static $k = 5$ keeps five passages readable and supplies the remaining evidence through the trained latent path. We reproduce that operating point and vary explicit context, then investigate textual allocation under a separate frozen reader. Our adapted Mistral SARA specialist is therefore retained explicitly, while its scores are not treated as matched Qwen evidence-budget comparisons.

2.2 Granularity and token-level compression

Dense X Retrieval [8] investigates retrieval units from passages to finer propositions. It supports examining the unit of evidence rather than treating a passage boundary as fixed. Its generated, self-contained propositions differ from our exact source sentences and rule-derived clause spans; our results do not establish that sentence or proposition granularity is universally best. LLMLingua-2 [9] learns token retention through data distillation and a bidirectional encoder. It offers a token-level alternative to our passage/sentence fidelity menu. We qualified the smaller official mBERT checkpoint, but did not complete a matched controlled quality run; qualification is not evidence of comparative effectiveness.

2.3 Budget allocation and dynamic programming

Self-Correcting RAG [10] formulates context selection with multiple-choice, multidimensional knapsack constraints and describes Pareto-pruned DP, together with a separate NLI-guided tree-search generation stage. This is closely related optimization prior art. Our QBGF uses one additive evidence-token constraint and mutually exclusive fidelities of the *same source*,

with a fixed reader and no MCTS/NLI generation stage. Its DP is exact for the specified additive surrogate and serialization; that guarantee is neither an optimum of downstream F1 nor a novelty claim about DP itself. BEA instead recomputes marginal values as evidence grows and does not claim a global knapsack optimum.

2.4 Adaptive strategies and learned resource value

Adaptive-RAG [11] chooses retrieval and generation strategies according to question complexity. Our context-size controllers examine the narrower question of allocating readable evidence, while BEA-V2 estimates sentence helpfulness from historical leave-one-out observations.

2.5 MiniLM cross-encoder reranking and relevance versus utility

The conventional `cross-encoder/ms-marco-MiniLM-L6-v2` checkpoint [12] jointly scores a query–text pair for MS-MARCO relevance. It differs from the `all-MiniLM-L6-v2` bi-encoder used in earlier semantic routing, which embeds texts separately. Our exploratory reranker packs scored sentences under the same evidence budgets. Its strong SciFact point estimates and weaker HotpotQA results make it an important comparison, while its scores remain relevance estimates, not probabilities that a sentence will improve reader quality. Lost in the Middle [4] further motivates measuring what the reader can use, rather than equating more context or higher similarity with better answers.

Position of this project. Our work connects query-conditioned source representations, exact evidence-token accounting, and downstream utility analysis through matched empirical comparisons. Existing compression, reranking, and knapsack methods provide the methodological context; the reported result tables contain measurements from our own experiments.

3 Problem Formulation

Let a query be q and a retriever return candidate passages $\mathcal{D} = \{p_1, \ldots, p_K\}$. A reader receives evidence string E with the query and generates y. The evidence must satisfy

$$|\operatorname{Tok}(E)| \le B. \tag{1}$$

The ideal but unavailable choice would maximize downstream quality directly:

$$E^*(q) = \arg\max_E \operatorname{Quality}(\operatorname{Reader}(q, E)) \quad \text{s.t.} \quad |\operatorname{Tok}(E)| \le B. \tag{2}$$

Because answer quality is unknown before generation, practical allocators must optimize observable surrogates such as query–evidence similarity, retrieval score, redundancy, or predicted utility.

Budget and evidence conventions. B is an input-evidence ceiling, not an output-token limit or the entire chat prompt. No-evidence Qwen (NONE) is query-only. In SARA, $k = 0$ removes explicit passages but can retain latent evidence; it is not a no-retrieval baseline. Retrospective oracles inspect gold outcomes and are not deployable policies.

3.1 Dataset characteristics and evidence-budget motivation

We distinguish the original benchmark populations, the subsets used in our experiments, and the retained retrieved-pool sample. Statistics for one population are not used as estimates for another. All lengths below use the frozen Qwen3-8B tokenizer without truncation; source lengths exclude reader instructions and answers.

Table 1: Official dataset inputs, before project filtering. QASPER parentheses give distinct papers. SciFact uses one shared 5,183-abstract corpus. Public HotpotQA fullwiki test is a different context setting, not distractor test.

Dataset	Train	Validation / dev	Public test inputs
HotpotQA	90,447	7,405 distractor	7,405 fullwiki
QASPER	2,593 (888)	1,005 (281)	1,451 (416)
SciFact	809	300	300

HotpotQA lengths describe unique supplied context paragraphs, identified by title and exact sentence-list hash, not complete Wikipedia articles. QASPER lengths assemble the supplied title, abstract, section headings and paragraphs, not PDF layout or table cells. SciFact lengths describe full corpus abstracts; cited-document subsets and transitive claim groups are distinct. Public HotpotQA fullwiki and SciFact test inputs lack gold answers/labels; HotpotQA distractor test is unavailable locally and is not substituted. These descriptive source measurements do not constitute a new test evaluation.

Split-wise quantiles, query lengths, source associations, and preprocessing conventions are detailed in Appendix E. Table 3 describes the validation cohort used in the project. The reserved confirmation cohort is separate: 200 HotpotQA queries

Table 2: Original source lengths in Qwen tokens: full official TRAIN sources for HotpotQA/QASPER and the shared SciFact corpus. Units differ across datasets; these are not retrieved-pool or selected-context lengths.

Dataset	Unit	Count	Mean	Median	P90
HotpotQA	Supplied paragraph	483,696	115.8	104	195
QASPER	Supplied full paper	888	5,515.6	5,164	8,320.7
SciFact	Full abstract	5,183	309.0	282	477.8

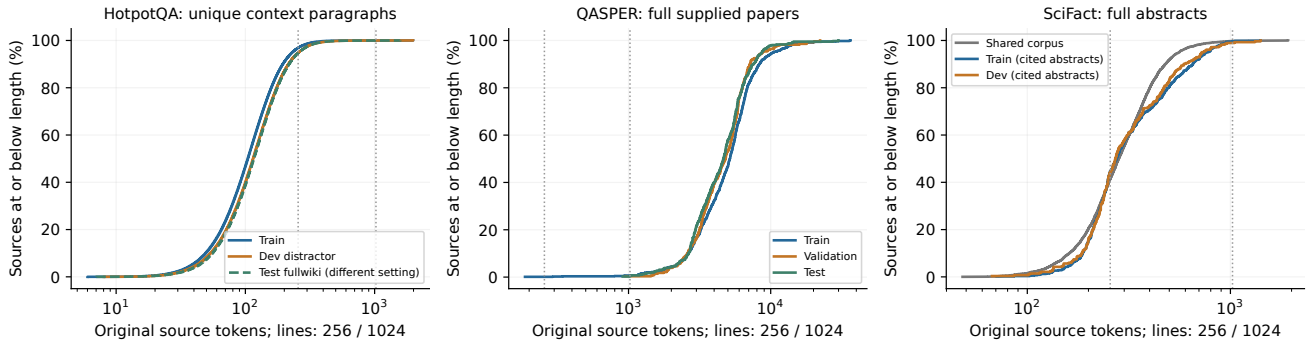


Figure 2: Original supplied source lengths by official split (Qwen tokenizer; log scale). The 256/1024 reference lines motivate selection but do not imply that entire documents enter the reader. SciFact cited subsets are shown separately from the shared corpus; HotpotQA test is fullwiki.

(200 groups), 50 SciFact claims (43 groups), and no fresh QASPER queries. These memberships are not combined when computing descriptive distributions or outcome metrics.

Table 3: Project validation cohort: 8,337 examples. Groups are the units used to prevent related examples crossing partitions; they are not necessarily individual documents.

Dataset	Examples	Source/group definition	Task / primary metric
HotpotQA	7,155	7,155 question groups; provided distractor context	Multi-hop QA / Answer F1
QASPER	1,005	281 distinct papers / paper groups	Scientific QA / Answer F1
SciFact	177	187 cited document IDs; 157 transitive cited-document components	Claim verification / macro-F1

Measurement and coverage. For a query, let L_q be the token cost of its entire retrieved candidate pool *before allocation*. We apply the frozen Qwen3-8B tokenizer and the experimental `additive-token-id-blocks-v1` serializer, including source-block separators. Query text, instructions, chat framing, and generated answers are excluded. This is neither full-document length nor the evidence actually selected by a method. Budget pressure is L_q/B, and the fraction requiring selection is $N^{-1}\sum_q \mathbb{1}[L_q > B]$.

Retrieved-pool length statistics are available for **450 retained SFR top-10 candidate pools (150 queries per dataset)**. The complete candidate pools for the full validation cohort are unavailable, so these descriptive statistics apply only to this exploratory subset, not to the full validation or confirmation populations. Pools with fewer than ten passages are counted without padding; this analysis did not rerun retrieval or reader generation.

The measured pools usually require selective retention even at 1024 tokens. Their passage sizes also differ: the mean standalone passage length is 138.2 tokens for HotpotQA, 247.7 for QASPER, and 332.2 for SciFact. QASPER passages are bounded at 256 tokens, whereas native paragraphs/abstracts in the other tasks can be longer. Thus both source construction and task contribute to the distributions. These observations motivate comparing full passages, selected sentences, and adaptive mixtures; they do not establish which method performs best. Pressure measures text burden, not the amount of useful information discarded.

3.2 Why granularity changes the problem

A whole passage preserves discourse and coreference but may spend most of a tight budget on surrounding text. A sentence is cheaper and can be combined with evidence from several sources, but can lose local context. Figure 4 illustrates this trade-off.

Table 4: Pre-allocation candidate-pool lengths in the retained exploratory sample ($n = 150$ per dataset). Pressure is the median L_q/B; exceedance percentages count each query once. Appendix D gives additional quantiles and passage/sentence statistics.

Dataset	Mean tokens	Median tokens	P90 tokens	> 256 (%)	> 1024 (%)	Median pressure $B = 256$	Median pressure $B = 1024$
HotpotQA	1,374.9	1,423.0	1,787.5	99.3	82.7	5.56×	1.39×
QASPER	2,458.3	2,512.0	2,567.0	100.0	100.0	9.81×	2.45×
SciFact	3,321.8	3,091.5	4,523.1	100.0	100.0	12.08×	3.02×

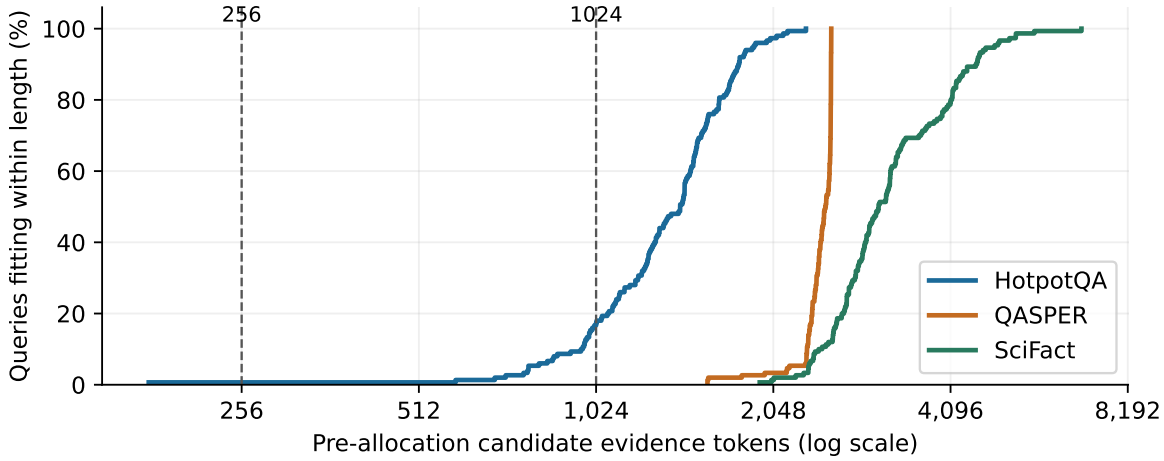


Figure 3: Empirical CDF of retrieved-pool token cost for 150 retained exploratory queries per dataset. Dashed lines mark the 256- and 1024-token evidence budgets. The curve height at a budget is the percentage whose entire pool fits. The horizontal scale is logarithmic; curves do not describe full source documents.

4 How the Hypothesis Evolved

4.1 SARA and adaptive readable context

SARA mixes k explicit text passages with $n - k$ compressed latent representations. This made k a natural control variable: a difficult query could receive more explicit text, while an easier query could rely more heavily on compressed evidence.

4.2 Static SARA $k = 5$: the parent baseline

The central starting baseline is **static SARA** $k = 5, n = 10$: five retrieved passages remain explicit and the remaining retrieved evidence follows the learned compression interface. The early QASPER development cohort comprises 231 questions from 86 papers drawn from a reformatted version of the official training split. With BM25 retrieval and the adapted Mistral/SARA reader, this baseline achieves **0.3629 Answer F1** and 0.1861 EM, with 1,157 mean context tokens in the recorded accounting. These are local project measurements, not the SARA paper's published benchmark score.

Later, the specialist-baseline extension evaluates the same fixed $k = 5, n = 10$ specialist setting on the 1,005-query official-validation-based QASPER cohort. Table 5 keeps both scales visible. The SFR arm changes the retriever, not k; neither specialist is constrained to the Qwen study's 256/1024 evidence-token policy. Their adapted reader, latent projector, prompt, and tokenizer differ from the final controlled text-only reader.

Table 5: Locally recorded static SARA $k = 5, n = 10$ baselines. Rows from different cohorts are not paired comparisons. All use the adapted Mistral specialist; none is a matched Qwen-256/1024 condition.

Project cohort	**Retriever**	**Queries**	**Answer F1**	**EM**
Early QASPER development (86 papers)	BM25	231	0.3629	0.1861
QASPER validation (281 papers)	BM25	1,005	0.3950	0.2100
QASPER validation (281 papers)	SFR	1,005	0.4366	0.2239

The 231-query value anchors the context-size response below; the 1,005-query values are separately scoped specialist references for the later validation study. They must not be pooled with the three-dataset Qwen table or used to infer a compression-only causal advantage.

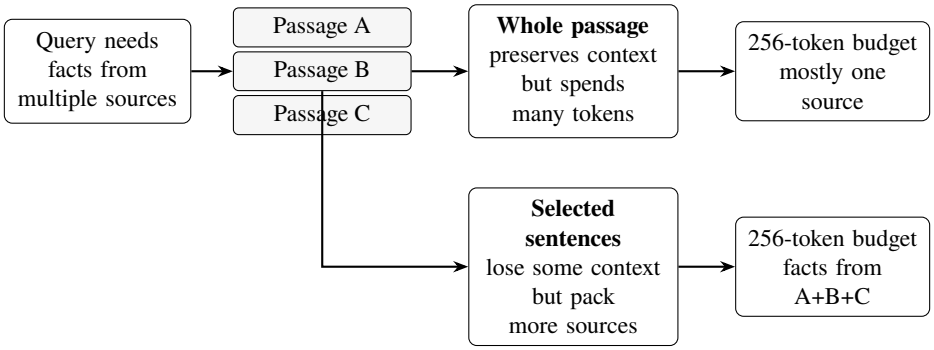


Figure 4: Granularity changes what can fit under the same budget. Fine units improve packing density, while whole passages preserve local discourse.

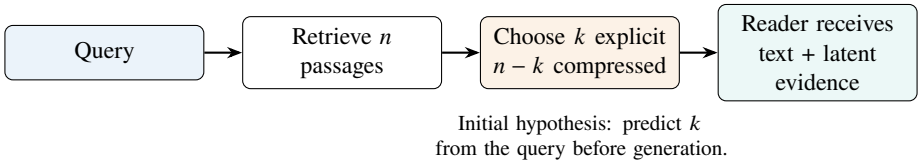


Figure 5: SARA-inspired starting point. The original adaptive variable was the number of explicit text passages, not evidence granularity.

4.3 Static context response and oracle headroom

We evaluated $k \in \{0, 2, 4, 5, 6, 8, 10\}$. Performance improved with more readable context but flattened after moderate k. A retrospective oracle, which selects the best k only after observing generation quality, showed substantial headroom.

The parent static $k = 5$ achieved 0.3629 F1; static $k = 8$ achieved 0.3950 F1 on the same historical cohort. The retrospective SARA oracle over $k \leq 8$ reached 0.5008 F1 (+26.78% relative). A cost-aware oracle retained 0.4894 F1 at mean $k = 3.49$, using 841.3 tokens versus 1,758.5 at $k = 8$ (52.16% fewer). Also, 46.32% of queries had $k = 0$ match or beat $k = 8$, and 77.92% had $k = 5$ match or beat $k = 8$.

Interpretation boundary. These oracle results demonstrate heterogeneity among tested choices; they do not show that the correct k can be inferred from an unseen query.

Concrete case pair: opposite context needs inside the same paper

Two questions from the same QASPER paper behave in opposite ways. One requires the sixth retrieved passage and rises from 0.1333 F1 at $k = 0$ to 0.6000 at $k = 6$; the other is best at $k = 0$ and degrades as explicit passages are added. Appendix A.7.1 gives the full questions, gold answers, retrieved evidence, generated answers, token counts, and per-k scores.

4.4 Why context-size prediction failed

We engineered 64 pre-generation features spanning BM25 score structure, query complexity, query–passage semantic alignment, passage redundancy/diversity, cumulative lexical coverage, and multi-aspect evidence signals. No feature remained significant for oracle-k association after Benjamini–Hochberg FDR correction at $q = 0.10$. Flat controllers therefore tended to save tokens by under-allocating evidence: logistic regression fell to 0.3034 F1 at mean $k = 2.03$; tree and rule policies reached only 0.3282–0.3286.

A sequential controller decomposed allocation into $2 \rightarrow 4 \rightarrow 6 \rightarrow 8$ decisions. The first transition was essentially unpredictable (ROC-AUC 0.4385), the middle $4 \rightarrow 6$ transition was more learnable (ROC-AUC 0.6503), and the final $6 \rightarrow 8$ transition offered little marginal gain. Nonlinear models did not solve the problem. Balanced logistic regression achieved 0.3842 policy F1 with 64.44% recall for useful $4 \rightarrow 6$ expansions; XGBoost reduced false expansions but also reduced recall to 46.67% and fell to 0.3725 F1.

The central difficulty was an **asymmetric decision cost**. Capturing a genuinely useful expansion produced an average gain of +0.3395 F1, whereas an unnecessary expansion incurred an average penalty of only −0.1345 F1—a 2.52× magnitude difference. In other words, a false stop can remove the one passage containing the decisive fact, while a false expansion often adds only mild distractor noise. A controller optimized primarily for classification accuracy or context reduction is therefore

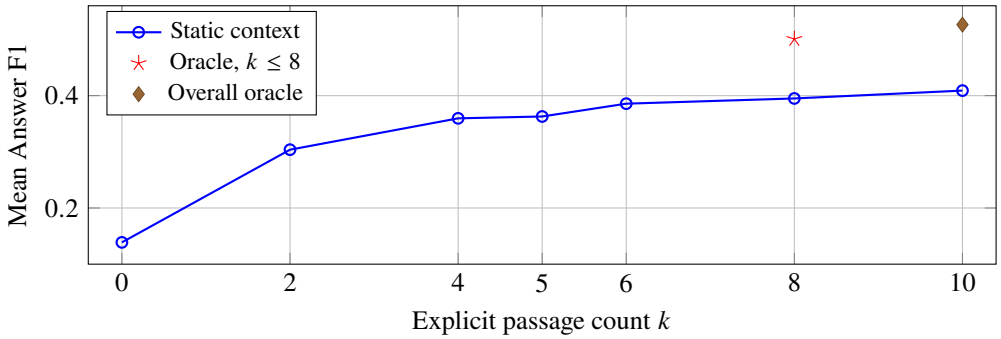


Figure 6: Static context-size response on QASPER. The oracle is an outcome-selected ceiling, not a deployable policy.

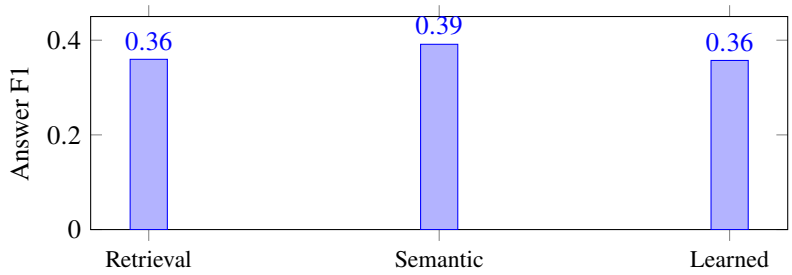


Figure 7: Passage identity at fixed $k = 4$ on the historical QASPER development cohort ($N = 231$); metric: Answer F1. The number of passages is unchanged; selecting different passages changes answer quality.

poorly aligned with the downstream objective: even a modest false-stop rate can dominate its end-to-end regret.

This asymmetry explains why increasingly complex controllers did not translate into better QA quality. Conservative policies saved tokens by predicting "stop," but they also missed expensive evidence transitions; policies tuned to preserve recall drifted back toward using nearly all available context. A content-aware controller ultimately matched static $k = 8$ quality (0.3952 vs. 0.3950) only at mean $k = 7.85$, yielding just 1.72% token savings. The result is not simply that the classifiers were weak; the *decision problem itself* penalizes under-allocation much more severely than mild over-allocation.

Concrete case: why under-allocation is unusually expensive

In one QASPER example, stopping at $k = 2$ omits the passage that explicitly names the gold categories and yields F1 = 0; expanding to $k = 4$ exposes that sentence and yields F1 = 1. A second example shows only a small loss from unnecessary expansion. Appendix A.7.2 reproduces the questions, gold answers, included/missing evidence, generated answers, and score changes.

First pivot. A scalar k hides which passages are retained and what information each contains. We therefore moved from evidence *quantity* to evidence *identity*.

4.5 Passage identity matters at fixed quantity

Holding the candidate pool and number of explicit passages fixed, we compared retrieval-order selection, dense semantic selection, and a learned passage-ranking model. At $k = 4$, semantic selection reached 0.39133 F1 versus 0.35960 for the retrieval-order prefix and 0.35721 for the learned selector.

Concrete case: same $k = 4$, different evidence identity

A controlled fixed-k example makes the identity effect concrete: all policies expose four passages, but only the route containing the lower-ranked passage with the English–German dataset description reaches the exact answer. Appendix A.7.3 shows the question, candidate passages, selected ranks, token counts, generated answers, and scores. This case illustrates the mechanism; aggregate results still show that the learned router did not beat semantic routing overall.

Additional routing, draft promotion, and extraction experiments. Earlier, separately scoped development studies tested whether an additional learner or an intermediate reader draft could improve evidence selection. Learned utility routing trailed the semantic $k = 8$ reference (0.39223 versus 0.40104 Answer F1). In the draft-promotion study, always exposing the promoted route reached 0.41257 versus 0.38906 without promotion, while a learned gate reached 0.40612; the gate did not surpass unconditional promotion. Query-only and draft-aware *generative* extraction scored 0.36866 and 0.37694, below their own no-intervention reference of 0.38563. Exact query-ranked source sentences reached 0.39002 in that extraction study, but its paired interval for improvement crossed zero. These are distinct experimental cohorts and pipelines, so their scores must not be compared directly with the final QBGF validation table. Appendix 18 records the local references, interventions, and outcomes. Collectively, the results favor preserving original source text and studying evidence selection directly rather than assuming that an additional learned gate or generative rewriting step will help.

4.6 Root-cause checks before changing granularity

Before attributing earlier behavior to allocation alone, we separated retrieval, representation, and reader-interface confounds. At retrieval depth 10, any-evidence recall was 0.618 for BM25, 0.636 for a MiniLM bi-encoder, and 0.727 for SFR-Embedding-Mistral. Candidate-pool quality was therefore materially different across retrievers. When retrieval was held fixed using the same SFR pool at $k = 6$, full-text RAG reached 0.41988 F1 and SARA's text-plus-latent mixture reached 0.42129; the paired bootstrap interval crossed zero. Thus, this controlled comparison did not isolate latent compression as the source of the earlier SARA advantage.

Why SARA latent compression was not simply transferred to every reader. SARA's compressed evidence is not a reader-agnostic vector store. Its implementation injects retriever vectors at `<COMPRESS>` token positions after a learned two-layer projector,

$$z_i = P_\phi(e_i), \qquad P_\phi : \mathbb{R}^{4096} \to \mathbb{R}^{4096}, \tag{3}$$

implemented as Linear(4096, 4096)–GELU–Linear(4096, 4096). The projector and LoRA adapter were trained against `Mistral-7B-Instruct-v0.2`; consequently, the projected vectors are learned for the Mistral hidden representation space and checkpoint interface. Even if another reader happens to have the same nominal hidden width, reusing the trained projector unchanged would not establish compatibility with that model's representation geometry. A Qwen- or Llama-based latent SARA system would require a newly compatible projection/adaptation path and validation rather than a drop-in reader swap. For the later cross-reader study we therefore compared textual evidence representations directly instead of claiming that the Mistral-trained latent projector transferred across model families.

Why Qwen3-8B was selected as controlled reader. The exploratory reader panel also exposed an evaluation-interface problem on SciFact. Under the frozen exact-label response contract, Mistral produced 1,849 invalid strict-format outputs out of 1,850 conditions and Llama produced 1,826/1,850 invalid outputs, whereas Qwen produced zero invalid outputs. Explanations appended after the required label were scored invalid by the fixed parser. These strict scores remain valid measurements of the frozen interface, but they are heavily confounded as comparisons of claim reasoning. Qwen was therefore retained as the primary control for the subsequent granularity and allocation studies because it followed the required verification output format reliably; this is *not* a claim that Qwen is universally the best reasoning model.

Concrete case: a reader-interface confound, not a reasoning ranking

A SciFact example has gold label `SUPPORT`. The decisive abstract sentence supports TCR–CD3 involvement in immune-synapse maturation, but the evaluator accepts only a single exact label token. Qwen obeys that contract; Mistral and Llama return semantically correct labels wrapped in explanatory text and are therefore rejected. Appendix A.7.4 reproduces the claim, decisive evidence, raw outputs, and parser decisions.

These checks motivated a standardized study: fix retriever, reader, prompt contract, and budget, then vary only how candidate evidence is represented and selected.

Historical comparability. Scores in the SARA/QCCA exploration used earlier retrieval, reader, and adaptation configurations. They motivate the research direction but are not directly comparable to the final SFR + Qwen3-8B controlled study.

5 Controlled Multi-Granularity Evidence Study

The controlled pilot compared three extractive representations under identical candidate retrieval: **whole passages**, **sentences**, and **fine-grained source spans**. The standardized setup used SFR-Embedding-Mistral retrieval and Qwen3-8B as the princi-

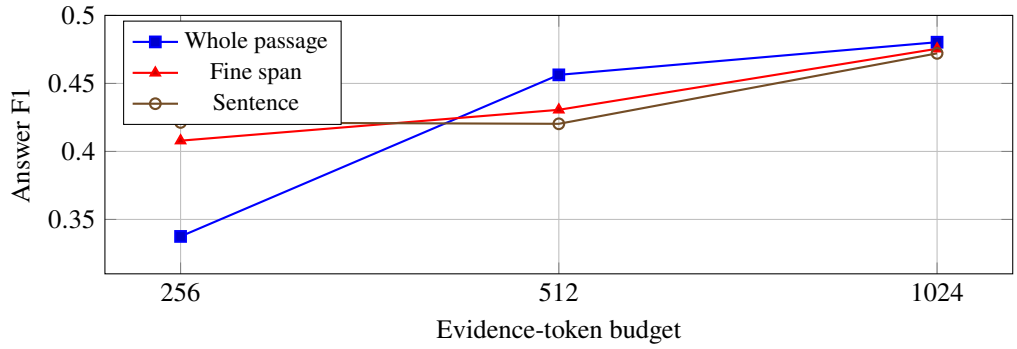


Figure 8: Granularity and budget in the HotpotQA pilot ($N = 50$). Sentences lead at 256 tokens, while whole passages have the highest point estimates at 512 and 1024 tokens.

Table 6: Controlled multi-granularity pilot on HotpotQA. Values are descriptive development-pilot estimates.

	256		512		1024	
Representation	Tok.	F1	Tok.	F1	Tok.	F1
Whole passage	165.9	0.3375	389.1	**0.4563**	798.4	**0.4803**
Fine extractive span	198.3	0.4079	426.7	0.4306	865.0	0.4755
Sentence	201.2	**0.4213**	423.4	0.4203	866.7	0.4721

pal reader. The question was whether the preferred evidence unit changes with available space.

At 256 tokens, sentence selection improved F1 by +0.0838 over whole passages, a +24.8% relative gain. A plausible explanation is that smaller units distribute a tight budget across more sources, but the pilot did not isolate source coverage as the causal mechanism. At 1024 tokens, whole passages were slightly higher, consistent with the possibility that preserved local context becomes more useful when space is less scarce. Formal reader-by-granularity interaction intervals crossed zero, so this pilot does not establish a universal crossover law.

Design hypothesis: rather than choose one granularity globally, choose a representation separately for each retrieved passage under the current token budget.

6 QBGF: Query-Conditioned Budgeted Granularity Fusion

6.1 Method overview

QBGF operates after retrieval and before generation. For each of the top-$K = 10$ retrieved passages, it constructs a small menu of mutually exclusive representations: `DROP`, the top 1/2/4/8 query-relevant sentences, or the `FULL` passage. A global optimizer then chooses exactly one option per passage subject to the evidence-token budget.

6.2 Step 1: query-conditioned passage importance

Let $s(p_i) = \cos(\mathbf{e}_q, \mathbf{e}_{p_i})$ be SFR cosine similarity between query and passage p_i. QBGF converts these scores into global passage relevance mass:

$$R_i = \frac{\exp(\tau_p s(p_i))}{\sum_{m=1}^{K} \exp(\tau_p s(p_m))}, \qquad \tau_p = 5. \tag{4}$$

Thus R_i expresses how much of the retrieved pool's relevance mass is assigned to passage i.

6.3 Step 2: relevance mass inside each passage

Each passage is segmented into rule-based sentences $\mathcal{S}(p_i) = \{s_{i,1}, \ldots, s_{i,n_i}\}$. Sentence cosine scores are normalized within the passage:

$$m_{i,j} = \frac{\exp(\tau_s s(s_{i,j}))}{\sum_{\ell=1}^{n_i} \exp(\tau_s s(s_{i,\ell}))}, \qquad \tau_s = 5. \tag{5}$$

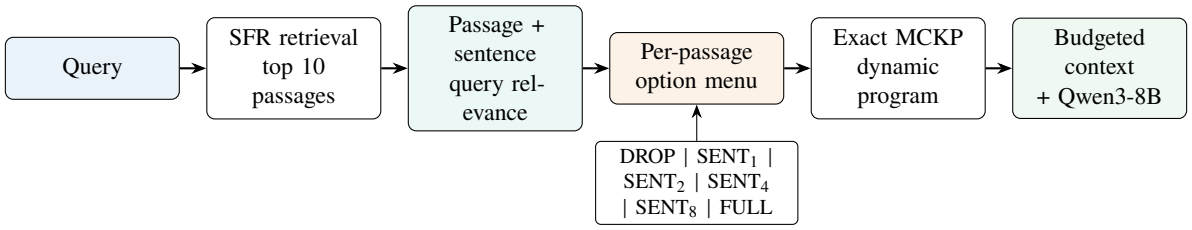


Figure 9: QBGF in one picture. The method generates no new evidence; it chooses among extractive representations already present in the retrieved passages.

For the top-k sentence option,

$$M(i,k) = \sum_{j \in \text{top-}k} m_{i,j}, \qquad u(i, \text{SENT}_k) = R_i M(i,k). \tag{6}$$

The full passage receives $u(i, \text{FULL}) = R_i$; dropping it receives zero.

6.4 Step 3: exact cost and mutually exclusive representations

For every passage,

$$O_i = \{\text{DROP}, \text{SENT}_1, \text{SENT}_2, \text{SENT}_4, \text{SENT}_8, \text{FULL}\}. \tag{7}$$

Costs are measured by serializing the actual evidence block and tokenizing it with the Qwen tokenizer, including separators and formatting. For example,

$$c(i, \text{SENT}_k) = |\text{Tok}(\text{Serialize}(\text{top-}k \text{ selected sentences}))|\,. \tag{8}$$

The full-passage cost is defined analogously on the serialized full block. This distinction matters: the constraint applies to the prompt that is actually passed to the reader, not to an approximate sum of unformatted sentence lengths.

6.5 Step 4: exact multiple-choice knapsack optimization

Each passage contributes exactly one option; `DROP` represents omission. The resulting multiple-choice knapsack problem is

$$\max_{o_1,\ldots,o_K} \quad \sum_{i=1}^{K} u(i, o_i) \tag{9}$$

$$\text{s.t.} \quad \sum_{i=1}^{K} c(i, o_i) \le B, \qquad o_i \in O_i. \tag{10}$$

An integer dynamic program solves the surrogate exactly:

$$V_i(c) = \max_{o \in O_i,\, c(i,o) \le c} \left[V_{i-1}(c - c(i,o)) + u(i,o) \right]. \tag{11}$$

The cost is exactly additive because each parent block is encoded separately, including its trailing double newline; the reader receives the concatenated block token IDs. It is not the length obtained by retokenizing the concatenated display string. DROP costs zero. Query/instruction/chat framing and answer tokens are excluded. Ties within 10^{-12} use lexicographic option indices; final states prefer utility rounded to 12 decimals, then lower cost and lexicographic path. Selected blocks and sentences are restored to source order. Global optimality is for this finite surrogate and numerical tolerance.

Important distinction. The dynamic program is globally optimal for the defined relevance-surrogate objective. It is *not* globally optimal for downstream Answer F1 or macro-F1; the experiments test whether the surrogate is good enough in practice.

Worked allocation case: QBGF avoids a lexical distractor

A saved 256-token QASPER query asks “What was the baseline used?” The pool contains both a high-overlap sentence about *manual annotation* and the decisive comparison to the Rei (2016) model. QBGF retains the latter and identifies the error detection system by Rei2016 (F1 = 0.8000), while the matched sentence and cross-encoder alternatives fail to name that system (F1 = 0). Appendix A.7.5 shows the evidence choices, token use, raw outputs, and five-method comparison.

6.6 Baselines and variants: what each comparison tests

All methods operate on the same retrieved candidates and budgets.

- **No evidence**: isolates the reader's parametric performance.
- **Whole-passage baseline**: scans passages in retrieval rank order, includes a full passage if it fits, skips a non-fitting passage and continues, then restores source order.
- **Sentence baseline**: segments every candidate, globally ranks sentences by query cosine similarity, greedily packs fitting sentences using exact serialized cost, and restores source order.
- **Hierarchical hybrid**: uses sentence units for the first three retrieved passages and finer exact source spans for the remaining seven, followed by the historical prefix allocator. It is a control, not a candidate eligible for final promotion.
- **No-query QBGF**: sets passage weights to $R_i = 1/K$ and removes query-conditioned sentence preference, while the upstream SFR candidate pool remains query-dependent. This tests the value of conditioning *inside the allocator*.
- **Passage-only QBGF**: restricts each menu to {DROP, FULL}; this tests the DP and passage weighting without sentence decomposition.
- **Sentence-only QBGF**: restricts each menu to DROP plus SENT_1/SENT_2/SENT_4/SENT_8; this tests whether the FULL option itself adds value.

A no-mass variant and greedy-density heuristics were considered during formulation but were not evaluated across the full validation matrix, so they are not used as primary evidence.

7 Experimental Setup

7.1 Datasets, tasks, and metrics

We use three complementary datasets so that evidence allocation is tested on both general-domain and scientific text, and on both answer generation and claim verification.

HotpotQA: combining facts across sources. HotpotQA contains questions whose answers can require linking facts from multiple Wikipedia paragraphs [5]. We use the standard *10-paragraph distractor setting*: the candidate material is the question's provided context, not a search over all of Wikipedia. Supporting-fact annotations identify relevant statements for retrospective evaluation. The allocation challenge is to retain both the bridge fact and the answer-bearing fact when they occur in different sources.

QASPER: answering questions about research papers. QASPER pairs information-seeking questions with scientific papers, answer references, and annotated supporting evidence [6]. A short answer may depend on a small methodology or results sentence within a much longer paper. The project adapts the retained paper text into candidate passages and finer source units; questions from the same paper must be treated as related examples when defining groups and uncertainty.

SciFact: checking a scientific claim. SciFact associates claims with scientific abstracts and evidence annotations [7]. In the project's normalized three-class interface, the reader returns exactly `SUPPORT`, `REFUTE`, or `NOT_ENOUGH_INFO`, depending on whether the supplied evidence supports, contradicts, or does not establish the claim. This task tests polarity-sensitive evidence selection as well as strict output compliance. Native SUPPORT/CONTRADICT annotations map to SUPPORT-/REFUTE; claims with a present but empty evidence annotation map to NOT_ENOUGH_INFO. Missing public-test labels remain unlabelled, not NEI. The reader receives the claim and retrieved corpus text, not gold cited IDs as an answer hint. Invalid strict-format outputs count as incorrect, and macro-F1 is computed over all three classes, not averaged per-claim F1.

Answer F1 is primary for HotpotQA and QASPER; three-class macro-F1 is primary for SciFact. We never pool scores across these tasks. Gold references and evidence annotations support scoring and retrospective diagnostics; inference-time evidence selection does not receive the gold answer. Appendix B gives one recorded question or claim, source-evidence excerpt, and reference answer or label for each dataset. Table 7 summarizes the corresponding allocation challenges.

7.2 Data roles

We distinguish the historical development analyses from the validation data used for method selection. These experimental roles are distinct from the benchmarks' official train, validation, and test splits. The main validation matrix contains 7,155 HotpotQA queries, 1,005 QASPER queries (281 papers), and 177 SciFact examples (157 groups), for 8,337 total. The reserved confirmation contains 200 untouched HotpotQA examples and 50 SciFact examples (43 groups). QASPER is excluded from confirmation because an earlier exploratory extraction run had already consumed its test query IDs.

7.3 Final controlled pipeline

All final methods share SFR top-10 retrieval, budgets $B \in \{256, 1024\}$, and base `Qwen/Qwen3-8B` in FP16 precision with greedy decoding and no task-specific LoRA adaptation. Qwen was kept as the controlled reader because the reader comparison

Table 7: Concrete task examples used throughout the analysis. They are illustrative executed cases, not substitutes for aggregate evaluation.

Dataset	Example	Why allocation is difficult
HotpotQA	A question about a Denis Villeneuve film requires the chain *Enemy* → Javier Gullón → *Invader (Invasor)*.	The final answer depends on preserving a bridge fact across different sources; dropping one hop can produce a plausible but wrong answer.
QASPER	"What are the five different binary classification tasks?" is answered by a short task list inside a scientific paper.	A small answer-bearing span competes with large amounts of semantically related methodology and background.
SciFact	A claim asks whether TCR/CD3 microdomains are required for immune-synapse activation; the gold label is `SUPPORT`.	A decisive assertion may be short, polarity-sensitive, and evaluated under a strict categorical output contract.

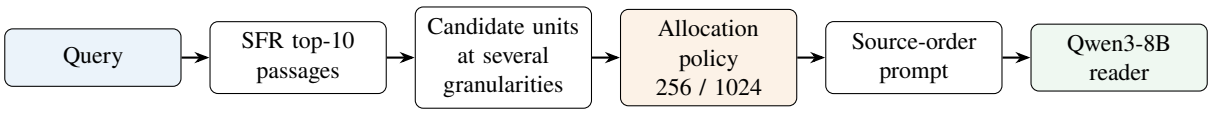


Figure 10: Standardized pipeline for validation and confirmation. Retrieval and reader inference are held fixed so comparisons isolate evidence allocation.

showed reliable compliance with the frozen SciFact exact-label format (zero invalid strict-format outputs in the reader panel), avoiding the severe parser-format confound observed for Mistral and Llama. The final study therefore standardizes the reader interface rather than claiming a universal reader ranking.

7.4 Pre-specified model-selection rule

The model-selection rule was fixed before evaluating the reserved confirmation cohort. QBGF would replace the sentence baseline only if it established noninferiority in *all six* dataset-by-budget validation cells and did not increase mean prompt-token cost in any cell, with at least one strict reduction. The noninferiority margin was 0.003 F1 for QA and 0.010 macro-F1 for SciFact. Paired uncertainty uses 10,000 cluster-bootstrap resamples with fixed seed 42; registered comparison families use Holm correction. Validation preserves the original 39-comparison registry and reports joint Holm adjustment over 51 comparisons when the 12 fusion ablations are included. Confirmation uses eight primary and four secondary prespecified contrasts, adjusted within their separate families. Bootstrap intervals are nominal percentile intervals, not simultaneous confidence bands. Noninferiority requires the lower paired interval bound to exceed the negative margin; failure is inconclusive, not proof of inferiority.

8 Results

8.1 Main validation surface

Table 8 reports the complete controlled surface. No method dominates every task or budget: sentence selection is especially strong on QASPER and SciFact at 256 tokens, while passage-based methods become more competitive at 1024 tokens.

8.2 QBGF ablation study: query conditioning and representation choice

To identify which design choices account for QBGF's performance, we compare the full allocator with three restricted variants on the *same frozen validation examples*, under the same retriever, reader, and evidence-token budgets. The no-query variant removes query-conditioned weighting *inside the allocator* (the SFR retrieval step remains query-dependent); the passage-only variant allows only `DROP` or `FULL`; and the sentence-only variant removes `FULL` while preserving the sentence-granularity menu. These restrictions isolate different design choices, although each may change both which sources are retained and the number of tokens used.

Query conditioning. Full QBGF outperforms its no-query variant in all six validation cells; these six contrasts are significant in the registered, Holm-adjusted comparison family ($p_{\text{Holm}} \leq 0.0114$). Since retrieval remains query-dependent in both conditions, this supports the contribution of *allocator-level* query weighting, not a query-independent retrieval comparison.

Representation choice. The nearly zero full-minus-sentence-only differences at 256 tokens show that admitting `FULL` adds no clear benefit at that budget. At 1024 tokens, full QBGF is below passage-only on HotpotQA (-0.0202) and below sentence-only on SciFact (-0.0243); these are descriptive point estimates, not evidence that either restricted policy is universally preferable. Passage-only performs substantially worse on SciFact at 256 (-0.2172 relative to full), consistent with the difficulty of packing whole abstracts into a tight budget. Unlike the no-query contrasts, the table alone does not establish statistical significance for the representation restrictions. The raw per-method scores are in Table 8, and the exact restrictions are

Table 8: Main validation surface. Values are Answer F1 for HotpotQA/QASPER and macro-F1 for SciFact. Bold is the highest point estimate in each row. Deltas use unrounded source values; subtracting rounded cells may differ by 0.0001.

Dataset	Budget	None	Passage	Sentence	Hybrid	QBGF	No-query	Passage-only	Sentence-only
HotpotQA	256	0.1653	0.4070	0.4082	**0.4253**	0.4121	0.3086	0.3935	0.4122
HotpotQA	1024	0.1653	0.4521	0.4233	0.4364	0.4323	0.4123	**0.4525**	0.4323
QASPER	256	0.0822	0.1693	**0.2634**	0.2544	0.2355	0.1473	0.1693	0.2356
QASPER	1024	0.0822	0.2880	0.2988	**0.2995**	0.2951	0.2284	0.2896	0.2977
SciFact	256	0.4207	0.4564	**0.7001**	0.6856	0.6750	0.4880	0.4578	0.6750
SciFact	1024	0.4207	**0.6990**	0.6704	0.6859	0.6604	0.5426	0.6031	0.6847

Table 9: QBGF component ablations on the full validation cohorts. Each entry is the full QBGF score *minus* the restricted-variant score, computed from the displayed four-decimal entries in Table 8. Positive values favor full QBGF. Metrics are Answer F1 for HotpotQA/QASPER and macro-F1 for SciFact. These descriptive differences are not additional hypothesis tests.

Dataset	Budget	− No-query	− Passage-only	− Sentence-only
HotpotQA	256	+0.1035	+0.0186	−0.0001
HotpotQA	1024	+0.0200	−0.0202	0.0000
QASPER	256	+0.0882	+0.0662	−0.0001
QASPER	1024	+0.0667	+0.0055	−0.0026
SciFact	256	+0.1870	+0.2172	0.0000
SciFact	1024	+0.1178	+0.0573	−0.0243

summarized in Appendix A.3.

These ablations support the use of query-conditioned evidence scoring but do not justify replacing the sentence baseline under the pre-specified, cross-dataset promotion criterion.

8.3 Why QBGF was not promoted

The final decision was not based on whichever method had the best point estimate in a particular row. Table 10 applies the pre-specified noninferiority rule. Only HotpotQA at 1024 tokens clears the quality margin; five cells fail to establish noninferiority.

The token-cost condition itself was satisfied: mean prompt-token differences for QBGF relative to the sentence baseline were −3.85, −8.17, −2.54, −2.54, −2.71, and −2.66 tokens across the six cells. Promotion therefore failed because of quality, not budget usage. The sentence baseline was locked as the final policy before confirmation was opened.

8.4 Reserved confirmation

The confirmation set tests the selected policy without reopening model selection. Table 11 reports all policies for transparency; the hierarchical hybrid remained a descriptive control rather than an eligible post-hoc replacement.

The selected sentence policy significantly improves over no evidence in all four confirmation cells. The clearest granularity result occurs on SciFact at 256 tokens: sentence selection reaches 0.7014 macro-F1 versus 0.4138 for whole passages, a +0.2876 absolute gain (Holm-adjusted $p = 0.0356$). At 1024 tokens, whole passages become competitive again. On HotpotQA at 1024, QBGF has a significant secondary gain over the sentence baseline (+0.0344 F1, Holm-adjusted $p = 0.0316$), but this cell-specific result does not identify which QBGF component caused the gain and does not overturn the locked selection protocol.

9 Why Relevance Optimization Is Hard: Leave-One-Out Attribution

If QBGF solves its knapsack exactly and query conditioning clearly matters, why does it still fail to dominate a simple sentence baseline? The likely bottleneck is the *objective being optimized*, not the dynamic program itself.

For selected evidence unit u_i in context E, we remove only that unit, keep the remaining selected evidence fixed, do not refill the freed budget, and regenerate:

$$\Delta_i = \text{Score}(E \setminus \{u_i\}) - \text{Score}(E). \tag{12}$$

For QA, Score is per-query Answer F1 with threshold $\delta = 0.01$: helpful evidence has $\Delta_i < -0.01$, harmful evidence $\Delta_i > +0.01$, and the rest is neutral. For SciFact, Score is per-example correctness and $\delta = 0$. Neutral therefore means no sufficiently large observed single-unit effect under this intervention; it does not imply that the unit is semantically irrelevant or ignored.

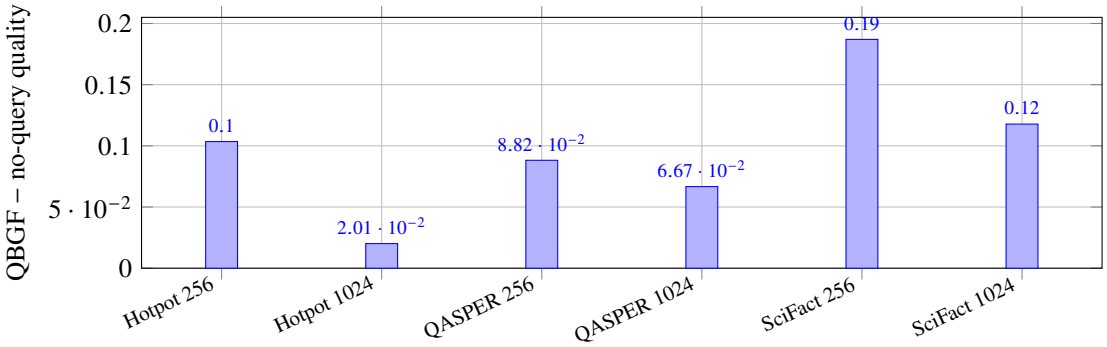


Figure 11: Effect of query conditioning within QBGF. Query-conditioned allocation is consistently better than the no-query variant, even though full QBGF does not consistently beat the strong sentence baseline.

Table 10: Noninferiority analysis for QBGF relative to the sentence baseline. Δ is QBGF minus sentence quality.

Dataset	Budget	Δ	95% CI	NI margin	NI?
HotpotQA	256	+0.0038	[−0.0035, +0.0110]	0.003	No
HotpotQA	1024	+0.0090	[+0.0045, +0.0135]	0.003	**Yes**
QASPER	256	-0.0279	[−0.0410, −0.0149]	0.003	No
QASPER	1024	-0.0037	[−0.0125, +0.0050]	0.003	No
SciFact	256	-0.0251	[−0.0878, +0.0365]	0.010	No
SciFact	1024	-0.0100	[−0.0711, +0.0505]	0.010	No

LOO is observable on individual evidence units

The aggregate neutral fractions are easier to interpret with actual interventions. Appendix A.7.6 gives three complete traces: a HotpotQA sentence whose deletion drops F1 from 1.0000 to 0.0870 (Δ = −0.9130), a second selected unit whose deletion changes nothing (Δ = 0), and a high-relevance SciFact distractor whose deletion flips an incorrect prediction to the correct `SUPPORT` label (Δ = +1). Each trace shows the evidence itself and the original/ablated reader behavior.

9.1 LOO findings and implications for allocation

The ablations in Table 12 reveal two recurring patterns. First, many selected units have no detectable *single-unit* effect in the tested context: 84.0% of HotpotQA QBGF units are neutral at 256 tokens, and 88.1% at 1024; the SciFact QBGF 1024 condition is 98.9% neutral. Second, the low relevance–deletion-effect rank correlations suggest that high semantic similarity alone is a weak proxy for downstream reader impact in these conditions. Neither finding proves that every neutral unit is dispensable: untested interactions between units, alternative evidence and output variability may obscure their contribution.

Qualitative inspection of 16 detailed query profiles suggested three recurring patterns: redundant evidence about the same entity relation, high-similarity distractors, and sentence fragmentation that removes antecedents/coreference. These are plausible explanations, not mechanisms proven by single-unit deletion. The high-relevance harmful SciFact example in Appendix A.7.6 is particularly instructive: removing one selected distractor corrects the output, despite that unit's semantic relevance.

How LOO connects to BEA-V2. The descriptive helpful/neutral/harmful fractions above use the stated $\delta = 0.01$ QA tolerance; BEA-V2 classifier supervision instead uses the strict sign of the saved deletion effect, $\Delta_i < 0$ versus $\Delta_i \geq 0$ (Section 10). These two definitions serve different analyses and must not be conflated. The supervised signal is therefore an *observed deletion effect in an existing context*, not a directly measured gain from adding the same unit to a new context.

10 From Relevance Packing to Utility-Aware Allocation: BEA

The leave-one-out findings motivate a shift in objective. QBGF already had an exact optimizer, yet many selected units had little measurable marginal effect on the reader. The problem was therefore not simply "solve the knapsack better." QBGF maximizes a non-negative relevance surrogate under a token ceiling and has no explicit downstream-utility-aware stopping rule. The budget is *not* mathematically required to be full, but additional semantically relevant options can continue to receive positive surrogate value even when their downstream marginal utility is neutral. This motivated a different allocation

Table 11: Reserved confirmation. The selected sentence policy is marked †; bold denotes the highest descriptive point estimate in each row.

Dataset	Budget	No evidence	Passage	Sentence†	Hybrid	QBGF
HotpotQA	256	0.1399	0.4227	0.4373	**0.4622**	0.4406
HotpotQA	1024	0.1399	0.4589	0.4329	0.4570	**0.4672**
SciFact	256	0.3686	0.4138	0.7014	**0.7067**	0.6870
SciFact	1024	0.3686	**0.7680**	0.7440	0.7168	0.6346

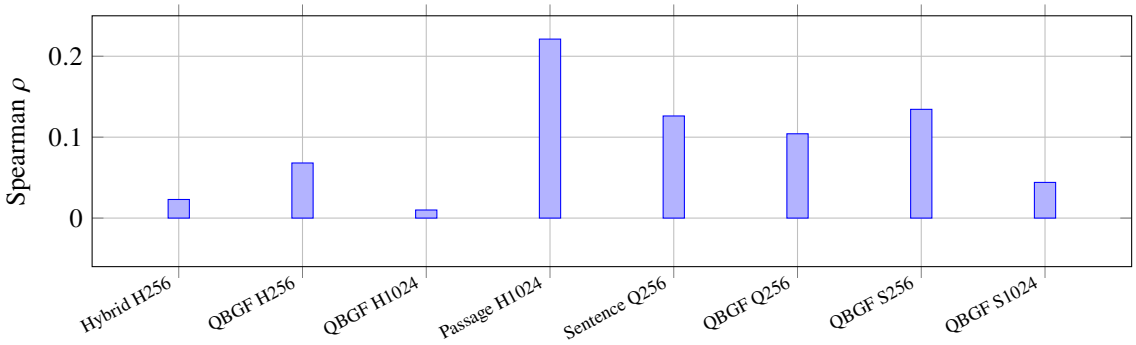


Figure 12: Dense relevance versus observed deletion utility on 50 developmental queries per dataset (question/paper/citation-component groups), at the displayed evidence budgets; H/Q/S denote HotpotQA/QASPER/SciFact. Most correlations are weak; the whole-passage HotpotQA 1024 condition is a notable positive exception.

philosophy: **start from little evidence and expand only while the next addition appears useful**.

10.1 BEA-V1: incremental evidence allocation with stopping

BEA-V1 (Budgeted Evidence Allocator V1) builds an evidence context *incrementally* from an empty selection, unlike QBGF's global multiple-choice knapsack. This process does not construct a document tree or merge source passages. At iteration t, let E_t denote the evidence already selected, and let a denote an *eligible transition*: adding one source sentence or replacing an accumulated source representation by its full passage when allowed. Write $T_a(E_t)$ for the context after applying transition a. The allocator evaluates several possible next transitions, chooses one, updates the context, and scores the remaining transitions again. A transition refers to the *change in selected evidence*, not to rerunning the language model at every step.

Exact semantic gain. The canonical research allocator uses normalized SFR embeddings; cosine values are clipped only to $[-1, 1]$, not to nonnegative relevance. Query aspects $A(q)$ are obtained by splitting on question marks, semicolons, newlines, commas and the words *and*, *but*, *versus*, *whereas*; phrases shorter than two words are discarded, duplicates are removed case-insensitively, and at most six remain. If none remains, the normalized whole query is used. This deterministic lexical decomposition is not learned semantic multi-hop decomposition.

Let S_t index the sentences exposed at state t, and let v_i and v_a denote normalized sentence and aspect vectors. The code implements

$$\begin{aligned} r_i &= \cos(e_q, v_i), \qquad h_a(S) = \max\big(0, \max_{j \in S} \cos(v_a, v_j)\big), \\ C_q(S) &= \sum_{a \in A(q)} h_a(S), \\ \Delta C_i(S) &= \sum_{a \in A(q)} \max\big(0, \cos(v_a, v_i) - h_a(S)\big), \\ \mathrm{Red}_i(S) &= \max_{j \in S} \cos(v_i, v_j). \end{aligned} \tag{13}$$

For the empty set, both h_a and redundancy are zero. Redundancy may be negative; it is a maximum over individual selected sentence vectors, not a context-mean embedding. Optimization uses the *sum* of aspect maxima. A separately reported diagnostic averages the aspect scores; the allocator itself uses their sum. A high aspect cosine is a similarity signal, not proof that the aspect has been answered.

V1 proposes every unselected sentence separately. A FULL replacement becomes eligible after two positive sentence additions from its parent. For FULL, the remaining parent sentences are scanned in source order: compute their marginal

Table 12: Representative leave-one-out functional distributions on the same 50-query-per-dataset developmental attribution cohort. Fractions count selected-unit interventions, not unique sentences.

Condition	ρ	Helpful	Neutral	Harmful
HotpotQA, QBGF, 256	0.068	8.9%	84.0%	7.2%
HotpotQA, QBGF, 1024	0.010	5.0%	88.1%	6.9%
QASPER, Sentence, 256	0.126	15.6%	76.4%	7.9%
QASPER, QBGF, 256	0.104	17.0%	69.5%	13.5%
SciFact, QBGF, 256	0.134	2.8%	92.2%	4.9%
SciFact, QBGF, 1024	0.044	0.5%	98.9%	0.6%

gain against a hypothetical growing set, accumulate only gains strictly above the context gate 0.05, and update that hypothetical set only for accepted sentences. The FULL value sums those accepted gains, while its actual representation exposes the entire passage. Already exposed sentences are excluded from this gain scan. Thus a compound transition is not assigned one pooled cosine or simply the sum of independent sentence gains.

Table 13: BEA-V1 scoring terms: what the allocator is trying to reward or penalize.

Term	How the value is obtained / interpreted	Effect on selection
$\mathrm{Rel}_t(a)$	Dense query–candidate similarity; for a single sentence the trace reports query–sentence cosine similarity.	Rewards query-matching evidence.
$\Delta\mathrm{Coverage}_t(a)$	Recalculate query-aspect coverage with and without the proposed transition; take the difference. Already-covered aspects contribute little new gain.	Rewards newly covered query information.
$\mathrm{Redundancy}_t(a)$	Semantic overlap between the proposed evidence and the context already selected, recalculated as the context grows.	Penalizes repetitions.
$\Delta c_t(a)$	Tokenize the *serialized evidence prompt* after and before the transition; subtract the two token counts.	Penalizes expensive additions.

From the components to the decision. BEA-V1 combines the three components with fixed, heuristic weights (not learned coefficients):

$$g_t(a) = \mathrm{Rel}_t(a) + 0.5\,\Delta\mathrm{Coverage}_t(a) - 0.5\,\mathrm{Redundancy}_t(a). \tag{14}$$

The 0.5 multipliers give partial credit to newly covered aspects and a partial penalty to overlap, while keeping direct relevance as the anchor. To compare candidate transitions of different lengths, BEA divides this gain by the number of *additional serialized tokens*:

$$\Delta c_t(a) = \left|\mathrm{Tok}(\mathrm{Serialize}(T_a(E_t)))\right| - \left|\mathrm{Tok}(\mathrm{Serialize}(E_t))\right|, \qquad d_t(a) = \frac{g_t(a)}{\Delta c_t(a)}. \tag{15}$$

Thus $g_t(a)$ measures the *estimated benefit of adding evidence*, whereas $d_t(a)$ measures the *estimated benefit per extra token*. Fitting positive-gain transitions are eligible; positive incremental cost uses the displayed ratio, while zero or negative incremental cost receives priority (the code uses a large sentinel). The algorithm chooses $a_t^* = \arg\max_a d_t(a)$ among eligible candidates, accepts it if the configured stopping/expansion conditions are met, and otherwise stops even when budget remains. Exact serialization includes double-newline source separators. Source identifiers are metadata, not inserted labels; headings count only when already present in source text. Query text and reader/chat framing are outside the evidence budget.

Worked calculation from a saved multi-hop trace. Consider the sentence linking screenwriter Javier Gullón to his Goya nomination for *Invader (Invasor)*. After other evidence had already been chosen, the BEA-V1 trace assigned this 56-token addition relevance 0.805983, new query-aspect coverage 0.095912, and redundancy 0.757633. Substituting the *recorded values* into Eqs. (14)–(15) gives

$$\begin{aligned} g_t(a) &= 0.805983 + 0.5(0.095912) - 0.5(0.757633) \\ &\approx 0.475123, \\ d_t(a) &= 0.475123/56 \approx \mathbf{0.008484} \quad \text{per token.} \end{aligned} \tag{16}$$

This example illustrates an important failure mode: even a *highly query-relevant* bridge sentence can receive a low gain-per-token score when earlier selections have made it appear redundant and it is relatively long. It was not selected in the recorded

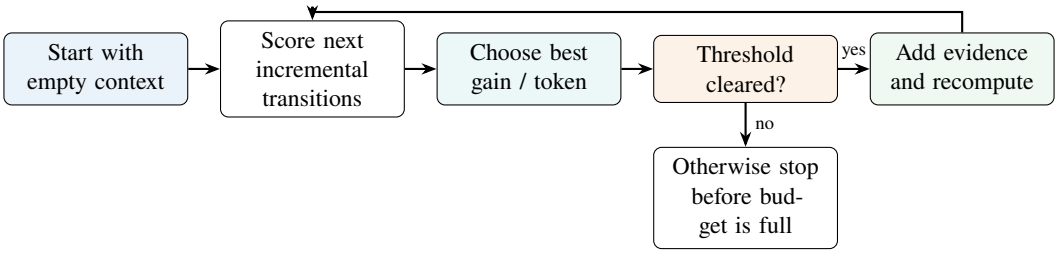


Figure 13: Incremental evidence allocation in BEA. Unlike QBGF's one-shot relevance-surrogate knapsack, BEA recalculates marginal values after each accepted transition and can stop before exhausting the evidence budget.

V1 case, and the answer was wrong (see Appendix A.7.7). The scores are computed *before generation*; the resulting Answer F1 is used only later to evaluate the policy.

V1 adds one sentence at a time, with FULL replacements as above; discrete $1 \to 2 \to 4 \to 8$ source states belong to V2, not V1. Candidate gains are recalculated after each accepted choice. The evaluated BEA-V1 configuration uses stop and expansion thresholds of 0.01 and a local context gate of 0.05, chosen using allocation-only calibration on nine development queries (18 query-budget cases). Candidates must meet their individual density threshold before the highest-density admissible transition is selected; ties use gain, incremental cost, source order, then transition kind. Appendix A.2 summarizes the three allocator designs side by side.

What BEA-V1 exposed. The intended benefit was to avoid spending tokens merely because budget remained. The observed failure mode was the opposite extreme: *context starvation*. On the matched HotpotQA cohort at the 1024-token budget, BEA-V1 used only 284 evidence tokens on average and achieved 0.3441 F1, below the sentence baseline at 0.3893. A local heuristic threshold can reject an addition whose immediate gain looks weak even when later evidence would be complementary. The experiment establishes under-allocation as a concrete failure mode; it does not prove that any single threshold decision caused every wrong answer.

Concrete starvation case: BEA-V1 stops before the answer-bearing region

On a 1024-token QASPER query asking for five binary classification tasks, BEA-V1 stops after only 245 evidence tokens and never reaches the task list; the reader instead returns stimulus words and scores F1 = 0. The matched sentence and QBGF contexts expose the task list and score 0.7826. Appendix A.7.7 gives the query, gold answer, observed allocation, raw output, token usage, and matched baselines.

10.2 BEA-V2: learn the probability that an evidence unit helps

BEA-V1 ranked candidate additions using the fixed heuristic in Eq. (14), but sometimes stopped before essential evidence was selected. BEA-V2 retains the *same* definitions of relevance, incremental coverage, redundancy, and exact token cost; its change is to weigh the non-negative heuristic gain by a prediction of *how likely a candidate is to help the reader*. This is a learned weighting of the next transition, not a replacement for the coverage/redundancy calculations or an additional generation step.

From LOO measurements to P_{help}. The leave-one-out (LOO) experiment in Section 9 removes one selected evidence unit, regenerates the answer, and measures the change $\Delta_i = \text{Score}(E \setminus \{u_i\}) - \text{Score}(E)$. For classifier supervision, an evidence unit is *helpful* if deleting it strictly decreases the recorded per-query score:

$$y_i^{\text{help}} = \mathbb{1}[\Delta_i < 0], \qquad y_i^{\text{harm}} = \mathbb{1}[\Delta_i > 0]. \tag{17}$$

Thus, a helpful sentence receives target label 1; neutral and harmful sentences receive label 0 for this particular binary prediction problem. For SciFact supervision, the score is per-claim correctness (0/1), not a per-claim macro-F1. The historical attribution data provide 8,537 usable sentence observations: 555 helpful, 552 harmful, and 7,430 neutral.

For each dataset, an L_2-regularized logistic classifier is trained on the available *pre-generation* candidate features $\mathbf{x}_a$—features describing the query and candidate evidence before the reader produces an answer. The trained model estimates

$$P_{\text{help}}(a) = \Pr(y^{\text{help}} = 1 \mid \mathbf{x}_a) = \sigma(\mathbf{w}_h^\top \mathbf{x}_a + b_h), \quad \sigma(z) = \frac{1}{1+e^{-z}}. \tag{18}$$

A larger P_{help} therefore means that the model assigns a higher probability to the historical LOO helpfulness label; it is *not* an observed F1 gain or a guarantee of causal usefulness for every future context. Features were standardized within training

folds, and five-fold grouped cross-fitting kept queries or document groups out of their own training folds (HotpotQA: query; QASPER: paper; SciFact: cited-document component). The classifier used $C = 1.0$ and the `lbfgs` solver. The saved pipelines use StandardScaler and L2 logistic regression without class weighting or imputation (maximum 1,000 solver iterations; seed 42). All 12 feature values must be finite floats. The original training labels use the strict sign, not the 0.01 tolerance used in some descriptive HELP/HURT summaries. At evaluation, each query uses its single assigned held-out fold model, not an ensemble or all-data refit; missing models or unassigned queries fail rather than silently falling back. Each of the fifteen dataset/fold models is associated with disjoint training and held-out groups.

Classifier evidence, distinct from answer quality. The 8,537 observations repeat some sentences across methods/budgets; they are not 8,537 independent sentences. They come from 50 developmental queries/groups per dataset, overlapping the exploratory answer cohort and main validation population. Group cross-fitting excludes each evaluated query/group from its predictor's training, but neither removes development reuse nor supplies independent confirmation. There is no reserved-confirmation query overlap. Table 14 uses pooled saved out-of-fold probabilities (rather than unweighted means of fold AUCs). The metrics are calculated from recorded out-of-fold probabilities; no model refitting was performed.

Table 14: Helpful-label classifier, pooled out-of-fold diagnostics. AP is average precision, not trapezoidal PR area. Brier is lower-is-better. Classification uses a diagnostic threshold of 0.5, not a tuned allocation cutoff.

Dataset	Rows	Helpful	Prev.	ROC-AUC	AP	Brier	Recall@.5
HotpotQA	2,941	188	.0639	.6249	.1064	.0593	.0000
QASPER	3,543	332	.0937	.5897	.2024	.0828	.0452
SciFact	2,053	35	.0170	.8361	.0721	.0167	.0000

An always-negative classifier already attains 93.5% accuracy overall. Its balanced accuracy is 0.5; the dataset prevalence is the descriptive no-discrimination AP reference. Fold-training-prevalence Brier baselines are .0601/.0862/.0168 respectively. These comparisons support some ranking signal but limited probability discrimination, especially in the QA tasks; they do not establish well-calibrated probabilities or accuracy of a future addition. Precision/recall/F1, confusion matrices and all features are given in Appendix F.

How the learned probability changes allocation. BEA-V2 multiplies BEA-V1's non-negative heuristic gain by the predicted helpfulness probability:

$$U_t(a) = P_{\text{help}}(a)\ \max\big(0,\ \text{Rel}_t(a) + 0.5\,\Delta\text{Coverage}_t(a) - 0.5\,\text{Redundancy}_t(a)\big)\,. \tag{19}$$

The ingredients are therefore calculated in two stages. First, BEA-V1's semantic and coverage routines compute $g_t(a)$ from the *current* selected context. Second, a previously trained classifier converts the candidate's pre-generation feature vector $\mathbf{x}_a$ into $P_{\text{help}}(a) \in [0, 1]$. A larger predicted probability retains a greater fraction of the same nonnegative heuristic gain; a smaller probability discounts it more strongly. Because $P_{\text{help}} \le 1$, weighting does not increase a candidate's absolute heuristic value, but can change the ordering of competing transitions. The $\max(0, \cdot)$ term prevents negative heuristic gains from becoming desirable through multiplication. At inference time, neither gold answers nor actual LOO interventions are computed for the new query.

The transition is scored by learned value per added token,

$$D_t(a) = \frac{U_t(a)}{\Delta c_t(a)}, \tag{20}$$

where $\Delta c_t(a)$ is the actual incremental serialized token cost. The allocator repeatedly chooses the best eligible transition, updates coverage and redundancy, and continues or stops using the configured threshold. For a sentence i, the displayed utility is exact. For a compound transition, $U_t(a) = \sum_{i \in a} P_{\text{help}}(i) \max(0, g_i(S_{t,i}))$, with hypothetical state $S_{t,i}$ updated after each addition and candidates reranked by weighted value. There is no independently trained passage-upgrade probability. Sentence probabilities are precomputed per query/budget; coverage and redundancy change with the state. The next per-source sentence-count target is one, two, four or eight (capped by available sentences); FULL competes after two positive contributions. Unlike V1, V2 has no 0.05 local context gate. It requires positive total value, budget feasibility, and the global density threshold; nonpositive incremental costs receive priority. Harmfulness predictions are diagnostic only. The essential change from V1 is *how a candidate's estimated value is computed*; the sequential allocation and stopping framework remains in place.

Worked BEA-V2 calculation on the same decisive sentence. At BEA-V2's *first* selection step, the context is empty; the recorded trace for the same 56-token Goya-nomination sentence reports relevance 0.805983, incremental coverage 0.803187, redundancy 0, and predicted helpfulness $P_{\text{help}} = 0.125925$. The two scores are therefore

$$\begin{aligned} g_0(a) &= 0.805983 + 0.5(0.803187) - 0.5(0) = 1.207577, \\ U_0(a) &= 0.125925 \times 1.207577 \approx 0.152065, \\ D_0(a) &= 0.152065/56 \approx \mathbf{0.002715} \quad \text{per token.} \end{aligned} \tag{21}$$

The candidate was selected at Step 1 and helped preserve the two-hop chain leading to *Invader (Invasor)*. The V1 and V2 calculations use *different allocation states* (partly populated versus empty); their numerical ratios should not be compared as though only the probability factor changed. The instructive comparison is that V2 brought the bridge evidence into the prompt early, whereas V1 failed to retain it in this recorded case. This example illustrates one successful transition, not a universal guarantee for unseen queries.

BEA-V2 results at the examined stopping threshold. We evaluated BEA-V2 at $\tau = 8\times10^{-5}$ alongside the frozen sentence baseline, QBGF, BEA-V1, and a conventional MiniLM cross-encoder sentence reranker on matched development cohorts ($N = 50$ per dataset, at both budgets). At this threshold, BEA-V2 had the highest mean Answer F1 of the five methods on HotpotQA at *both* 256 tokens (**0.4352** versus 0.3661 for sentence selection) and 1024 tokens (**0.4234** versus 0.3893). In a worked multi-hop case, it selected an important linking sentence that BEA-V1 missed and recovered the correct answer (Appendix A.7.7). On QASPER and SciFact, other methods retained the highest point estimates, as shown in Table 15. The MS-MARCO `cross-encoder/ms-marco-MiniLM-L6-v2` provides the conventional reranking comparison.

This threshold was examined after development diagnostics rather than chosen by a successful independent calibration-and-confirmation procedure. The HotpotQA improvements are therefore *exploratory* and do not establish BEA-V2 as a validated replacement across datasets. They motivate further testing of learned helpfulness under a separately calibrated stopping rule.

Worked BEA-V2 case: learned helpfulness recovers a missing bridge

On the HotpotQA *Enemy* → *Javier Gullón* → *Invader* chain, BEA-V1 does not expose the decisive bridge and scores 0 F1. BEA-V2 assigns that sentence one of the highest P_{help} values, selects it at Step 1, and recovers the exact answer. Appendix A.7.7 reproduces the two evidence paths, the decisive sentence, token use, P_{help}, transition value, and outputs. This is a concrete example of the improved allocation behavior; the broader comparison remains exploratory.

Table 15: Exploratory matched-cohort comparison after the final policy had already been locked ($N = 50$ per dataset). Bold is the largest descriptive point estimate in each row. These results did not reopen model selection.

Dataset	Budget	Sentence	QBGF	BEA-V1	BEA-V2	MiniLM CE
HotpotQA	256	0.3661	0.4042	0.2976	**0.4352**	0.3404
HotpotQA	1024	0.3893	0.4020	0.3441	**0.4234**	0.3618
QASPER	256	**0.2668**	0.2641	0.1868	0.2618	0.2641
QASPER	1024	0.2896	**0.2964**	0.2084	0.2904	0.2827
SciFact	256	0.6289	0.5310	0.6127	0.5522	**0.6855**
SciFact	1024	0.5533	0.5881	0.6432	0.5772	**0.6479**

BEA-V2 has the highest HotpotQA point estimates at both budgets, exceeding sentence selection by +0.0691 F1 at 256 tokens and +0.0341 at 1024. Other methods lead on QASPER and SciFact. In a distinct 24-test family comparing MiniLM with each of four controls (sentence, QBGF, BEA-V1, BEA-V2) across six dataset-budget cells, no comparison survives Holm correction ($p_{\text{Holm}} \geq 0.2592$). This family does *not* test the BEA-V2-versus-sentence contrasts. Nominal intervals use 10,000 paired cluster-bootstrap resamples; permutation tests use cluster label swaps with seed 42. These exploratory results condition on the trained predictors and do not include model-fitting uncertainty or independent confirmation.

Scope of the learned-helpfulness comparison. BEA-V1 versus BEA-V2 is a *policy-level comparison*, not a clean ablation of P_{help} in isolation: the versions also differ in their gating and stopping configurations. No matched BEA-V2 run with $P_{\text{help}} \equiv 1$ and all other BEA-V2 settings held fixed is reported here. Consequently, the observed policy differences motivate

learned utility weighting but do not, on their own, establish how much of the improvement is caused solely by the predictor. A matched no-helpfulness ablation is a useful future test, not an already measured result.

Final methodological progression. QBGF asks, "Which representation best fits the evidence budget?" LOO reveals that retrieval relevance is not the same as downstream usefulness. BEA-V1 introduces incremental selection and stopping but sometimes stops too early. BEA-V2 learns to prioritize evidence from LOO helpfulness and achieves the best matched HotpotQA F1 at both tested budgets; broader task robustness remains an open question.

11 Discussion

11.1 What the project learned

Across these stages, five conclusions emerge:

1. **Context demand is heterogeneous.** Retrospective oracle choices vary substantially across queries.
2. **That demand is difficult to predict as a scalar budget.** Pre-generation context-size controllers are fragile under asymmetric under-allocation costs.
3. **Evidence identity and representation are first-class control variables.** Changing which passages survive matters even at fixed k, and granularity has budget-dependent descriptive patterns in the controlled pilot.
4. **Optimization quality is limited by objective quality.** QBGF solves its relevance-based MCKP exactly; ablations show that query weighting matters, but the relevance objective is only weakly aligned with observed reader utility in many conditions.
5. **Learning usefulness can help, especially on multi-hop QA.** BEA-V1 shows that stopping too aggressively can starve the reader of complementary evidence. BEA-V2 recovers useful evidence in several worked cases and has the highest matched-cohort HotpotQA F1 at both budgets, though the results are exploratory and other tasks favor other methods.

This explains why the simple sentence baseline is so competitive. It combines a strong dense relevance signal with a representation that packs efficiently under tight budgets, without adding another learned controller, a relevance-surrogate optimization layer, or a learned stopping rule that can amplify prediction error.

11.2 Negative results as design guidance

Several negative results materially narrowed the design space. Nonlinear context controllers sacrificed recall; learned passage ranking did not beat simple semantic routing; generative extraction degraded answer quality; latent compression was not isolated as the cause of the SARA advantage once retrieval was controlled; mixed granularity did not establish a consistent benefit over sentence-only QBGF; BEA-V1 starved the context under a large budget; and BEA-V2/MiniLM follow-ups did not establish a statistically defensible new winner. Together, these findings identify distinct failure modes: under-allocation, surrogate-objective mismatch, reader-interface sensitivity, and threshold calibration.

11.3 Limitations

- **Task scope:** HotpotQA uses its standard 10-paragraph distractor pool rather than unrestricted open-domain retrieval.
- **Reader specificity:** final controlled results use base Qwen3-8B in FP16. Qwen was selected as a controlled reader partly because it obeyed the frozen SciFact label interface; different readers may value context differently, and the reader panel itself was format-confounded on SciFact.
- **SARA transfer scope:** the trained SARA projector/LoRA pair is Mistral-specific. The project did not train a new latent projector for Qwen or Llama, so the final textual granularity study should not be interpreted as a cross-family test of SARA latent compression.
- **Pilot size:** the granularity pattern was discovered on a small controlled development pilot, and formal interaction intervals did not establish a universal effect.
- **Utility intervention and complementarity:** LOO measures the marginal effect of deleting one unit from one frozen context. In multi-hop settings, two units can be complementary: a sentence may appear neutral when its partner fact is already missing, yet become essential when the full reasoning chain is present. The observed neutral fraction therefore should not be read as intrinsic uselessness. Pair-deletion, Shapley-style attribution, or other interaction-aware analyses would better capture higher-order evidence effects.
- **QASPER confirmation:** final QASPER confirmation was deliberately omitted after earlier test-query exposure.

Runtime and deployment limits. Saved selections contain local `selection_seconds`, and some generations retain reader timing, but there is no matched end-to-end benchmark controlling cache state, preprocessing, hardware and concurrency across all three allocators. We therefore make no allocator-speed or serving-throughput ranking. Cached predictions remain conceptual reader calls; repeated LOO generations and classifier fitting are research costs, not free inference. Model-training

uncertainty, severe class imbalance, selected-context supervision and the nine-query failed stopping calibration limit deployment claims.

11.4 Future work

The experiments point to four concrete extensions rather than a broader search for more complex allocators. **First**, BEA-V2 threshold calibration should use a substantially larger, stratified development cohort while preserving the current rule that calibration cannot inspect reader F1, gold evidence, or held-out LOO labels. **Second**, QBGF should be stress-tested against its utility design choices, including the passage/sentence softmax temperatures (τ_p, τ_s) and the use of normalized within-passage sentence mass versus simpler unnormalized relevance scores. **Third**, downstream utility estimation should model evidence interactions—for example with pair deletion or interaction-aware attribution on multi-hop chains—rather than relying only on single-unit marginal effects. **Finally**, a systems evaluation should report allocator wall-clock overhead and token/latency savings relative to reader prefill and decoding, separating the cost of exact MCKP optimization or repeated BEA transition scoring from the much larger generation cost.

These extensions preserve the central lesson of the current study: the next improvement is more likely to come from a better estimate of *what actually helps the reader* and a better-calibrated stopping rule than from making the packing algorithm itself more elaborate.

11.5 Interactive Evidence Lab demonstration

We additionally implement *Evidence Lab*, a Streamlit interface for inspecting how evidence reaches the reader. Live Lab accepts a custom question and editable passages, retrieves a shared candidate pool, and compares FIXED_SENTENCE, QBGF, BEA-V1, and BEA-V2 under 256- or 1024-token *input-evidence* budgets. It exposes selected source units, token usage, and the actual reader answers. With a user-provided reference, the interface also supports bounded leave-one-out interventions: delete a selected unit, leave the freed budget unfilled, regenerate the answer, and inspect the original context, removed unit, remaining context, and score change. Run history supports controlled comparisons across settings.

The application is an explanatory implementation, not a separate benchmark or a change to the frozen final method. Its custom-input BEA-V2 uses a saved-model ensemble for demonstration, whereas the research evaluation uses a single held-out fold; implementation identity is not claimed. Reference agreement is not independent factual verification. Appendix C includes only the main comparison and deletion-analysis screenshots. Code and launch instructions are available through the GitHub repository linked on the title page.

12 Conclusion

This project began with a simple adaptive-context question inspired by SARA: *how much readable evidence should each query receive?* Query-specific headroom exists, but predicting one context size reliably is difficult. The project therefore shifted from quantity to evidence identity and then to representation. A controlled granularity pilot showed a descriptive shift: sentence-level selection led at 256 tokens, while full passages had competitive point estimates at larger budgets. The observed interaction was not statistically conclusive.

That observation motivated QBGF, which formalizes adaptive representation selection as an exact multiple-choice knapsack problem over query-conditioned passage and sentence options. The method validates an important idea—query conditioning matters—but it does not consistently beat a strong sentence baseline. A locked noninferiority rule therefore selects the simpler baseline for confirmation. Leave-one-out analysis then identifies the deeper problem: exact optimization of a relevance surrogate can still allocate evidence with little measurable marginal reader utility.

The final BEA branch follows directly from that diagnosis. BEA-V1 changes the question from "how do we fill the budget?" to "when is the next evidence increment worth adding?" but demonstrates the danger of context starvation. BEA-V2 estimates the probability of helpful evidence from historical LOO outcomes and uses it in the same incremental process. At the explored threshold $\tau = 8 \times 10^{-5}$, it achieves the highest HotpotQA point estimates at both 256 and 1024 tokens (0.4352 and 0.4234 F1), demonstrating the potential of learned utility-aware allocation. Because these results come from a development comparison and other datasets favor different methods, they do not overturn the previously selected sentence baseline. The project therefore ends with a sharper research target than where it began: **future adaptive RAG needs reliable, transferable estimates of downstream evidence utility and stopping decisions that avoid both irrelevant over-allocation and harmful under-allocation.**

Practical lesson: under strict evidence budgets, establish a strong extractive sentence baseline first. Add adaptive granularity or utility-aware stopping only when the utility signal is validated across tasks and the allocator is checked for both near-full waste and context starvation.

A Additional Experimental Evidence

A.1 Context-controller benchmark

This benchmark asks whether increasing controller complexity can recover the retrospective oracle headroom identified in Section 4. We report representative policies from the 4 → 6 expansion stage, where the predictive signal was strongest; because useful expansions have a substantially larger payoff than unnecessary ones, recall is particularly important when interpreting these results.

Table 16: Representative configurations from the 12-model 4 → 6 expansion benchmark on QASPER development data.

Model	Policy F1	Regret	Token red.	Recall	False exp.
Balanced logistic regression	0.3842	0.1053	36.54%	64.44%	28.57%
XGBoost	0.3725	0.1170	39.59%	46.67%	19.05%
Random forest	0.3717	0.1178	37.66%	55.56%	24.24%
HistGradientBoosting	0.3716	0.1179	37.86%	51.11%	24.68%
Decision tree	0.3671	0.1224	38.67%	35.56%	22.08%
MLP	0.3638	0.1257	38.57%	42.22%	20.78%
LightGBM	0.3632	0.1263	37.05%	48.89%	19.91%

A.2 Algorithmic specifications of the research stages

The three proposed allocators at a glance

QBGF: global representation allocation

For passage i, choose one

$$o_i \in \{\text{DROP}, \text{SENT}_{1,2,4,8}, \text{FULL}\}.$$

Query-conditioned utility is

$$M(i,k) = \sum_{j \in \text{top-}k} m_{ij}$$

$$u_k = R_i M(i,k)$$

with $u_{\text{FULL}} = R_i$. Then solve

$$\max_{\mathbf{o}} \sum_i u(i,o_i), \qquad \sum_i c(i,o_i) \le B.$$

The DP is exact for this surrogate.

How it works. QBGF starts with the full retrieved pool, builds a discrete fidelity menu for every passage, and chooses the globally best combination under the token ceiling. It decides *how to pack a fixed budget*; it has no downstream-utility-aware stopping rule.

BEA-V1: incremental heuristic stopping

At state t, score transition a:

$$g_t(a) = \text{Rel}_t(a) + 0.5\Delta\text{Cov}_t(a) - 0.5\text{Red}_t(a)$$

$$d_t(a) = \frac{g_t(a)}{\Delta c_t(a)}.$$

Choose

$$a_t^* = \arg\max_a d_t(a),$$

add it only if the registered stopping/expansion rules are cleared, then recompute all transitions.

How it works. BEA-V1 reverses the philosophy: begin with little evidence and grow the context one transition at a time. It can stop before the budget is full, which avoids automatic packing but creates the new failure mode observed in the project—context starvation.

BEA-V2: learned helpfulness

Cross-fitted LOO labels ($y = 1$ if deletion hurts the answer) train

$$P_{\text{help}}(a) = \sigma(w_h^\top x_a + b_h).$$

Primary value:

$$U_t(a) = P_{\text{help}}(a)\max(0, g_t(a)),$$

$$D_t(a) = \frac{U_t(a)}{\Delta c_t(a)}.$$

Choose the best next addition by helpfulness-weighted gain per added token.

How it works. BEA-V2 keeps BEA-V1's incremental allocation but weights the hand-designed relevance/coverage/redundancy gain by the predicted historical helpful-label probability. This learned value led the matched HotpotQA comparisons at both budgets, while performance varied across datasets.

Figure 14: Mathematical contrast among the three proposed allocation algorithms. QBGF optimizes a fixed relevance-based packing problem; BEA-V1 introduces incremental stopping; BEA-V2 weights the nonnegative heuristic by learned LOO helpfulness; compound transitions sum state-dependent sentence values.

BEA-V1 threshold and early stopping. The evaluated BEA-V1 configuration applies a marginal-density stopping threshold of 0.01, an expansion threshold of 0.01, and a local context gate of 0.05. In the QASPER starvation example, feasible positive-gain candidates remain, but the largest admissible density is $0.009965016 < 0.01$. The algorithm therefore terminates with reason `LOW_MARGINAL_VALUE`, having selected 245 of 1024 evidence tokens. This observed early stop illustrates the risk of rejecting complementary answer-bearing evidence.

BEA-V2 threshold calibration. An allocation-only calibration examined ten initial thresholds and four refinements near 8×10^{-5} on nine development queries (both budgets), without inspecting reader answers or gold labels. No setting satisfied

all pre-specified stopping and context-utilization constraints. The subsequent BEA-V2 comparison at 8×10^{-5} is therefore exploratory rather than a successfully calibrated or independently confirmed policy.

A.3 QBGF ablation design and comparison scope

The controlled QBGF variants alter the feasible representation menu or remove a component of its surrogate while keeping the SFR top-10 retrieval pool, Qwen3-8B reader, and evidence budgets fixed. The main results section reports quality differences; the following table states the precise experimental intervention.

Table 17: Interpretation of the four QBGF configurations in the controlled validation study; no additional reader executions are implied.

Configuration	Change to allocation	Question isolated
Full QBGF	Query-dependent passage weights; query-ranked sentence mass; `DROP`, sentence and `FULL` options.	Reference for all restricted variants.
No-query	Uniform passage weights and no query-conditioned sentence preference *inside the allocator*; retrieval is still SFR query-conditioned.	Does allocator-level query weighting help?
Passage-only	Restrict each source to `DROP` or `FULL`; retain query-dependent passage scoring.	How costly is excluding fine-grained sentence options?
Sentence-only	Allow `DROP` and sentence options, but remove `FULL`.	Does the whole-passage choice improve selection?

These are *representation/weighting ablations*, not changes to the underlying retriever or reader. Table 9 reports the differences and Table 8 contains the corresponding quality scores. The reported significance statement applies specifically to the no-query ablation comparisons, not automatically to the passage-only or sentence-only differences.

A.4 Supplementary routing, draft-promotion, and extraction results

Table 18 summarizes earlier development-stage comparisons. Each reference row denotes the baseline for its local study, not a shared benchmark across all stages.

Table 18: Intermediate experiments, with the algorithmic intervention and the conclusion that is actually supported by the result.

Study	Condition / algorithm	F1	What the number means
Passage identity ($k = 4$)	Retrieval-order prefix: keep the first four passages from the same candidate pool.	0.35960	Local reference. It fixes the number of readable passages and changes neither reader nor candidate pool.
	Semantic cosine routing: rerank the same candidates by $\cos(e_q, e_p)$ and expose the top four.	0.39133	A +0.03173 F1 point gain shows that *passage identity* matters even when passage count is fixed. It does not prove a universal dense-retrieval advantage.
	Learned passage ranking: expose the top four under a learned passage score $\hat{r}_\theta(q, p)$.	0.35721	The learned ranking condition did not improve over the simple semantic selector on this development cohort.
Learned utility routing	Semantic $k = 8$ reference: fixed dense semantic ordering.	0.40104	Strong simple reference for asking whether an extra utility-learning layer helps.
	Learned utility ranking: reorder candidates using the learned utility score $\hat{u}_\theta$.	0.39223	The additional learned ranker did not improve answer quality over fixed semantic ordering on this cohort.
Draft promotion	Base draft: answer from the base $k = 4$ route without promotion.	0.38906	Reference on the internal development cohort.
	Always promote: always expose the selected promoted route after the initial draft.	0.41257	Promotion can help, but it necessarily adds another generation/evidence stage. This is the best point estimate in this local study.
	Learned gate: use a fitted gate to decide whether promotion is needed.	0.40612	The gate improved over the base draft but trailed unconditional promotion, so it did not demonstrate that the promotion decision itself had been learned reliably.
Extraction	No intervention: retain the base evidence/draft pipeline.	0.38563	Reference on the larger fitting cohort.
	Query-only generative extraction: generate a compressed evidence text from query + source context.	0.36866	The generative extraction intervention reduced F1 by 0.0170 on this cohort. Qualitative review suggested incomplete extraction/syntax loss, but those mechanisms are not isolated causally.
	Draft-aware generative extraction: condition the extractor on query + current draft + source evidence.	0.37694	Also below the no-intervention reference; draft conditioning did not rescue the generative compressor.
	Exact query-ranked source sentences: select three exact source sentences instead of generating new text.	0.39002	A small +0.0044 point gain; the 95% CI $[-0.0085, +0.0175]$ crosses zero, so the result is suggestive rather than confirmatory.

A.5 Retrieval and reader confound checks

At depth 10 on the diagnostic cohort, any-evidence/all-evidence recall was 0.61818/0.43182 for BM25, 0.63636/0.44545 for the `all-MiniLM-L6-v2` bi-encoder, and 0.72727/0.52273 for SFR-Embedding-Mistral. Thus the later SFR candidate pool had materially higher evidence recall than the early BM25 pool.

Reader sensitivity was also substantial: under related controlled contexts, adapted C-Mistral scored 0.4071 F1 versus 0.2353 for public Mistral, 0.2738 for Qwen, and 0.2897 for Llama. These values alone do not define a universal reader ranking because prompts/adaptation differ. More importantly, the dedicated reader–granularity panel exposed a strict-format confound on SciFact: Mistral produced 1,849/1,850 invalid exact-label responses, Llama 1,826/1,850, and Qwen 0. Qwen was therefore retained as the controlled reader for later experiments to stabilize the response contract.

SARA's latent path was also model-specific. Its trained projector is a two-layer 4096-dimensional MLP feeding a Mistral-based model at `<COMPRESS>` positions, jointly used with a Mistral LoRA adapter. The project did not train an equivalent Qwen/Llama projector; consequently the final SFR+Qwen study deliberately compares textual evidence granularities rather than claiming cross-family SARA latent transfer.

A.6 Confirmation contrasts

For the selected sentence baseline versus no evidence, all four confirmation contrasts were significant after Holm adjustment: HotpotQA 256 +0.2974 F1 (95% CI $[0.2424, 0.3524]$, $p_{\text{Holm}} = 0.0008$); HotpotQA 1024 +0.2929 ($[0.2421, 0.3452]$, 0.0008); SciFact 256 +0.3328 macro-F1 ($[0.1386, 0.5142]$, 0.0126); and SciFact 1024 +0.3754 ($[0.1908, 0.5455]$, 0.0126).

Against whole-passage packing, the selected sentence policy had +0.0147 F1 on HotpotQA 256 (CI crosses zero), −0.0261

at 1024 (CI crosses zero), +0.2876 macro-F1 on SciFact 256 (95% CI $[0.0971, 0.4721]$, $p_{\text{Holm}} = 0.0356$), and −0.0239 at 1024 (CI crosses zero).

A.7 Detailed worked evidence cases

This appendix expands the compact examples referenced in the main text. Each case shows the task input, target, evidence actually exposed or omitted, downstream output, and score so that the mechanism behind the aggregate result is visible. Earlier examples retain the model configuration used at that stage and should not be compared numerically with the later controlled SFR+Qwen study unless the cohort is explicitly matched.

A.7.1 Example A.1: Opposite context needs within one QASPER paper

High-context example. Question: "Which psycholinguistic and basic linguistic features are used?"
Gold answer: Emotion Sensor Feature; Part of Speech; Punctuation; Sentiment Analysis; Empath; TF–IDF Emoticon features.
The top lexical passages repeatedly mention "psycho-linguistic features" and the shorthand "NLP Features," but they do not enumerate the requested feature set. The decisive methodology passage appears only at retrieval Rank 6 (BM25 score 1.337668):

> "Methodology ::: NLP Features — We have identified a novel combination of features ... The first one is the **Emotion Sensor Feature** ... The second one is the collection of selected topical signal from text collected using **Empath** ... We have borrowed the idea of using other features like **punctuation features** and **parts-of-speech tags** ..."

The recovered fixed-k execution trace shows a sharp threshold effect:

k	Context tok.	F1	Decisive R6?	Generated answer (abbreviated excerpts)
0	133	0.1333	No	"No linguistic features are used."
2	448	0.1429	No	"NLP Features"
4	826	0.1429	No	"NLP Features"
5	1084	0.1429	No	"NLP Features"
6	1326	**0.6000**	**Yes**	"Emotion Sensor Feature, Empath, Parts-of-speech tags, Punctuation features"
8	1735	0.4167	Yes	Longer feature list with extra distractor terms
10	2128	0.3478	Yes	Partial feature list with additional context

Thus the reader remains essentially stuck until Rank 6 enters the explicit context; adding still more passages then reduces F1 again. This is a concrete example of a non-smooth context requirement rather than a monotone "more context is better" effect.

Low-context example from the same paper. Question: "How have the differences in communication styles between Twitter and Facebook increase the complexity of the problem?"
Gold answer: "Systems do not perform well both in Facebook and Twitter texts."
At $k = 0$, the stored answer scores 0.2500 F1. Adding explicit retrieved text shifts the response toward generic domain-adaptation/code-mixing explanations and lowers the score; at $k = 10$ it is 0.2000. Two questions from the *same paper* therefore prefer opposite amounts of readable evidence.

Experimental setting. QASPER development examples evaluated with the SARA setup using Mistral-7B-Instruct-v0.2 and its learned latent projector.

A.7.2 Example A.2: QCCA false stop versus false expansion

False-stop case.
Question: "What are the near-offensive language categories?"
Gold answer: "inappropriate, discriminating."
At $k = 2$ (526 tokens), the two exposed passages discuss transfer learning and annotation challenges but do not name the categories. The reader answers "hate speech, discrimination, abusive language, offensive language" (F1 = 0). Expanding to $k = 4$ (1,071 tokens) exposes the decisive sentence:

> "Although not designed for offensive language detection, some categories are close to it, such as **inappropriate** and **discriminating**..."

The generated answer becomes "inappropriate, discriminating" (F1 = 1).

False-expansion contrast. The best stored $k = 0$ condition scores 0.2500 F1; adding two explicit passages ($k = 2$, 671 tokens) changes the response to a generic cross-domain explanation and scores 0.2000, only −0.05 lower. The pair illustrates why missing decisive evidence can be much more costly than carrying some extra context.

Experimental setting. QASPER development examples from the context-allocation experiments; the comparison varies only how much explicit evidence is exposed.

A.7.3 Example A.3: Evidence identity at fixed $k = 4$

Question: "What languages are seen in the news and TED datasets?"
Gold answer: "English, German."
All policies expose exactly four chunks from the same BM25 top-10 candidate pool. The decisive evidence is not in the lexical top four; it appears at Rank 8 (BM25 score 1.229125):

> "... we derive data from ParCorFull, an **English-German corpus** annotated with full coreference chains ... For our analysis, we select a portion of **English news texts and TED talks** from ParCorFull and translate them with the three NMT systems ..."

Ranks 1–5 repeatedly discuss news, TED talks, translation quality, error types, and coreference, but do not state the language pair. This is why high topical overlap is insufficient for the actual question.

Policy	Selected ranks	Tok.	Reader output	F1	EM
BM25 prefix	[1,2,3,4]	826	"English"	0.6667	0
Semantic cosine	[2,3,4,5]	991	"English, Spanish"	0.5000	0
Learned router	[1,2,6,8]	945	"English, German"	**1.0000**	**1**

Only the learned route brings Rank 8 into the four-passage budget and therefore exposes the exact English–German relation. This case demonstrates the *passage-identity mechanism*; it does not overturn the aggregate result in Figure 7, where semantic routing is stronger on average than the learned router.

Experimental setting. QASPER development example with the same top-10 candidate pool and the same four-passage budget for all three routing policies; the reader is the historical SARA/Mistral setup.

A.7.4 Example A.4: SciFact output-format confound

Claim: "TCR/CD3 microdomains are a required to induce the immunologic synapse to activate T cells."
Gold label: `SUPPORT`.
A decisive evidence sentence states:

> "Finally, by interfering with Nck-CD3 epsilon association in vivo, we demonstrate that TCR-CD3 recruitment of Nck is critical for maturation of the immune synapse and for T cell activation."

The evaluation parser permits exactly `SUPPORT`, `REFUTE`, or `NOT_ENOUGH_INFO`. The observed reader outputs are:

Reader	Raw output	Parser
Qwen	`SUPPORT`	VALID
Mistral	`SUPPORT` followed by an explanatory paragraph	INVALID
Llama	"Based on the provided abstract, the correct classification is: SUPPORT."	INVALID

The example shows why the panel is an interface-compliance diagnostic, not a clean ranking of biomedical reasoning ability.

Experimental setting. SciFact development example under the exact-label evaluation contract used in the reader comparison.

A.7.5 Example A.5: Worked QBGF allocation on a QASPER distractor case

Question: "What was the baseline used?"
Gold reference (verbatim): "error detection system by Rei2016". **Budget:** 256 evidence tokens.
The candidate pool contains text about manual annotation and adaptive learning rates as well as the decisive source sentence (verbatim):

> "For comparison, we also report the performance of the error detection system by Rei2016, trained using the same FCE dataset."

The saved matched outputs and token counts are:

Method	Tokens	Utilization	Reader output (verbatim)	F1
Sentence	256	100.0%	The baseline used was the available training set without any additional annotated data.	0.0000
QBGF	256	100.0%	The baseline used was the error detection system by Rei2016.	**0.8000**
BEA-V1	240	93.8%	AdaDelta was used as the baseline for calculating an adaptive learning rate during training.	0.0000
BEA-V2 post-hoc	255	99.6%	The baseline used was the available training set without any additional annotated data.	0.0000
MiniLM CE	256	100.0%	The baseline used was manual annotation.	0.0000

QBGF exposes the Rei2016 comparison sentence. The other observed answers instead refer to data availability, optimization or annotation; we do not infer the reader's hidden reasoning from those outputs. This case illustrates what joint representation allocation can recover, while the aggregate QASPER-256 result still slightly favors the sentence baseline; it is therefore an explanatory case, not a claim of universal QBGF superiority.

Experimental setting. QASPER matched comparison at a 256-token evidence budget with the controlled reader and retriever used in the final allocation study.

A.7.6 Example A.6: Three LOO interventions with the deleted unit shown

Helpful unit. A HotpotQA example asks, "What album released in 1998 features the song 'Risingson'?" (gold: *Mezzanine*). A selected 20-token unit with relevance 0.6605 states:

> "It is the first single from their third album "Mezzanine" and the eighth single overall."

With the unit present, the reader outputs "Mezzanine" (F1 = 1.0000); deleting only this unit reduces F1 to 0.0870, so $\Delta = -0.9130$ and the unit is classified HELPFUL.

Neutral unit. On the same query/context, the four-token title unit "Risingson" has relevance 0.5774. Deleting it leaves both output and F1 unchanged at "Mezzanine", 1.0000, hence $\Delta = 0$ and the unit is NEUTRAL. High similarity or inclusion in the selected context therefore does not imply measurable marginal contribution.

Harmful unit. A SciFact claim states:

> "Diabetic patients with acute coronary syndrome experience increased short-term and long-term risk for bleeding events."

The gold label is `SUPPORT`. One selected 40-token sentence has dense relevance 0.7882 (parent retrieval rank 1):

> "Because this disease may be associated with increased platelet aggregation, we investigated whether diabetic patients with ACS derive particular benefit from platelet glycoprotein (GP) IIb/IIIa receptor inhibition."

The full five-unit context (165 evidence tokens) makes the reader return `NOT_ENOUGH_INFO` (incorrect). Deleting only this sentence, without refilling the freed budget, leaves 125 evidence tokens and changes the output to `SUPPORT` (correct), giving $\Delta = +1.0000$: the unit is HARMFUL. Crucially, another selected sentence in the same original context explicitly states that patients with diabetes are "at increased risk of ischaemic events and bleeding post-acute coronary syndromes." This is an observed context-dependent prediction change despite on-topic similarity; the deletion does not establish an internal distraction mechanism.

Type	Intervention	Before	After	Δ
Helpful	Delete the Mezzanine album sentence	1.0000	0.0870	−0.9130
Neutral	Delete title "Risingson"	1.0000	1.0000	0
Harmful	Delete high-relevance ACS trial sentence	0	1	+1.0000

These are local interventions under frozen contexts: they measure contextual marginal utility, not intrinsic truthfulness or universal sentence quality.

Experimental setting. Leave-one-out interventions under the fixed reader: one selected unit is deleted, the freed budget is not refilled, and the answer is regenerated.

A.7.7 Example A.7: BEA-V1 starvation and BEA-V2 bridge rescue

BEA-V1 starvation on QASPER. A QASPER example asks:

> "What are the five different binary classification tasks?"

The gold answer lists presence/absence of consonants, phonemic nasal, bilabial, high-front vowels, and high-back vowels. The decisive 48-token source sentence states:

> "In order to explore the problem space, we thus specifically target five binary classification problems ... i.e. presence/absence of consonants, phonemic nasal, bilabial, high-front vowels and high-back vowels."

BEA-V1 never reaches this sentence. Its transition trace shows 15 accepted additions totaling only 245 evidence tokens; it then stops for `LOW_MARGINAL_VALUE`, with final marginal score 0.009965016 < 0.01 (the effective frozen stop threshold) and 779 of 1024 evidence tokens unused. The selected context instead contains stimulus words such as "pat, pot, knew and gnaw" and another unrelated token `INLINEFORM0`. The reader output was:

> "The five different binary classification tasks are pat, pot, knew, gnaw, and INLINEFORM0."

for F1 = 0. In contrast, FIXED and QBGF use 1022 and 1018 evidence tokens, include the decisive task-list sentence, and both return the correct list at F1 = 0.7826. The exploratory BEA-V2 run uses 965 tokens, also includes the decisive sentence, and reaches the same 0.7826 F1 on this query.

Method	Evid. tok.	Util.	Unused	F1	Outcome
FIXED	1022	99.8%	2	0.7826	Task list exposed
QBGF	1018	99.4%	6	0.7826	Task list exposed
BEA-V1	**245**	**23.9%**	**779**	**0.0000**	Stops before S005
BEA-V2 post-hoc	965	94.2%	59	0.7826	Task list recovered

This case makes the starvation failure concrete: a local incremental stopping rule can terminate while a large budget remains and before a complementary, answer-bearing unit is reached. The method-level V1 threshold configuration is reported separately in Appendix A.2.

BEA-V2 bridge rescue on HotpotQA. A HotpotQA multi-hop question requires the chain *Enemy* → screenwriter Javier Gullón → Goya nomination for *Invader (Invasor)*. The decisive saved sentence is shown below as an abbreviated excerpt (ellipsis marks omitted text):

> "He garnered a Canadian Screen Award nomination ... and a Goya Award nomination for Best Adapted Screenplay at the 27th Goya Awards for 'Invader (Invasor)'."

Before stopping, V1 has accepted 17 sentence units into a 168-token context, including short title units ("Javier Gullón", "Goya Award for Best Adapted Screenplay") and unrelated film details. Its largest remaining feasible density is 0.00958149, below the effective 0.01 threshold; the bridge itself has density 0.008484. The selected context and transition scores are reported in the corresponding experiment records. Thus the bridge does not simply lose a final ranking contest to an accepted alternative: after earlier cheap choices, it fails the stopping gate. BEA-V1 evaluates this sentence late, rejects it, uses 168 tokens, and the reader incorrectly answers that the nomination was for *Enemy* (F1 = 0). In the exploratory BEA-V2 run, the same bridge sentence receives $P_{\text{help}} = 0.125925$, is selected at Step 1 with value 0.152065 and ratio 0.002715, and the 243-token context yields "Invader (Invasor)" (F1 = 1). This demonstrates the intended behavior of the learned helpfulness signal on one case; it does *not* change the failed global calibration result or the mixed aggregate comparison.

Experimental setting. The QASPER example uses a 1024-token budget; the HotpotQA bridge example uses a 256-token budget. Both compare the evidence actually exposed to the same downstream reader under matched method conditions.

B Dataset Examples and Evidence Formats

The examples below are recorded development cases already used in the project, not new held-out results. They show the input, the kind of source evidence available to the system, and the required answer format. Longer method-level analyses are retained in Appendix A.7.

HotpotQA: a two-source reasoning chain

Recorded question. "What did the screewriter for a 2013 psychological thriller film directed by Denis Villeneuve get a Goya Award nomination for Best Adapted Screenplay at the 27th Goya Awards for?" The original spelling is preserved.

Evidence excerpts. One source identifies *Enemy* as the 2013 Denis Villeneuve thriller and names Javier Gullón as its writer. A second source states that he received "a Goya Award nomination for Best Adapted Screenplay at the 27th Goya Awards for 'Invader (Invasor)' (2012)."

Reference answer: *Invader (Invasor)*. The allocator must preserve the link from the film to its screenwriter and then to the nominated adaptation; retaining only the first hop can produce the plausible but wrong answer *Enemy*.

QASPER: a precise answer inside a scientific paper

Question. "What are the five different binary classification tasks?"

Source-evidence excerpt (citation gap marked by ellipsis). "In order to explore the problem space, we thus specifically target five binary classification problems … i.e presence/absence of consonants, phonemic nasal, bilabial, high-front vowels and high-back vowels."

Reference answer: presence/absence of consonants, phonemic nasal, bilabial, high-front vowels, and high-back vowels. A short list-bearing sentence answers the question; related background and stimulus words do not. The context-starvation example in Appendix A.7.7 shows why the reader must actually receive that sentence.

SciFact: a claim and a categorical decision

Recorded claim. "TCR/CD3 microdomains are a required to induce the immunologic synapse to activate T cells." The dataset wording is reproduced as recorded.

Source-evidence excerpt. "Finally, by interfering with Nck-CD3 epsilon association in vivo, we demonstrate that TCR-CD3 recruitment of Nck is critical for maturation of the immune synapse and for T cell activation."

Reference label: `SUPPORT`. The required output is the label itself. An explanation that contains the correct label can still violate the frozen exact-label parser; Appendix A.7.4 illustrates that distinction. Per-example correctness is not the cohort's three-class macro-F1.

C Interactive Evidence Lab Demonstration

Evidence Lab makes the project's evidence-allocation choices inspectable. The user supplies a query and passages, selects an evidence budget, and compares the four methods using the same retrieval and reader configuration. Each method card exposes its selected-evidence cost, answer, and optional reference-agreement score; exact source units can be expanded for inspection. The deletion view separately shows what the reader saw before removal, the removed unit, the unchanged remaining units, and the regenerated answer. The interface visualizes selected evidence and observed reader outputs, not the model's internal reasoning.

Figures 15 and 16 are actual UI captures from a bounded custom-preset run with SFR and Qwen3-8B. They are illustrative demonstrations, not additional validation or confirmation measurements. The World Cup examples use user-supplied passages and a supplied reference rather than benchmark annotations. The screenshots illustrate one custom-input execution; they do not establish benchmark performance or reproduce the frozen research allocators bit-for-bit. In particular the custom ensemble differs from the single-fold research predictor, and the demo QA tolerance differs from strict-sign classifier labels. BEA-V2 retains the exploratory status described in the main report. The GitHub link on the title page provides the application and setup instructions.

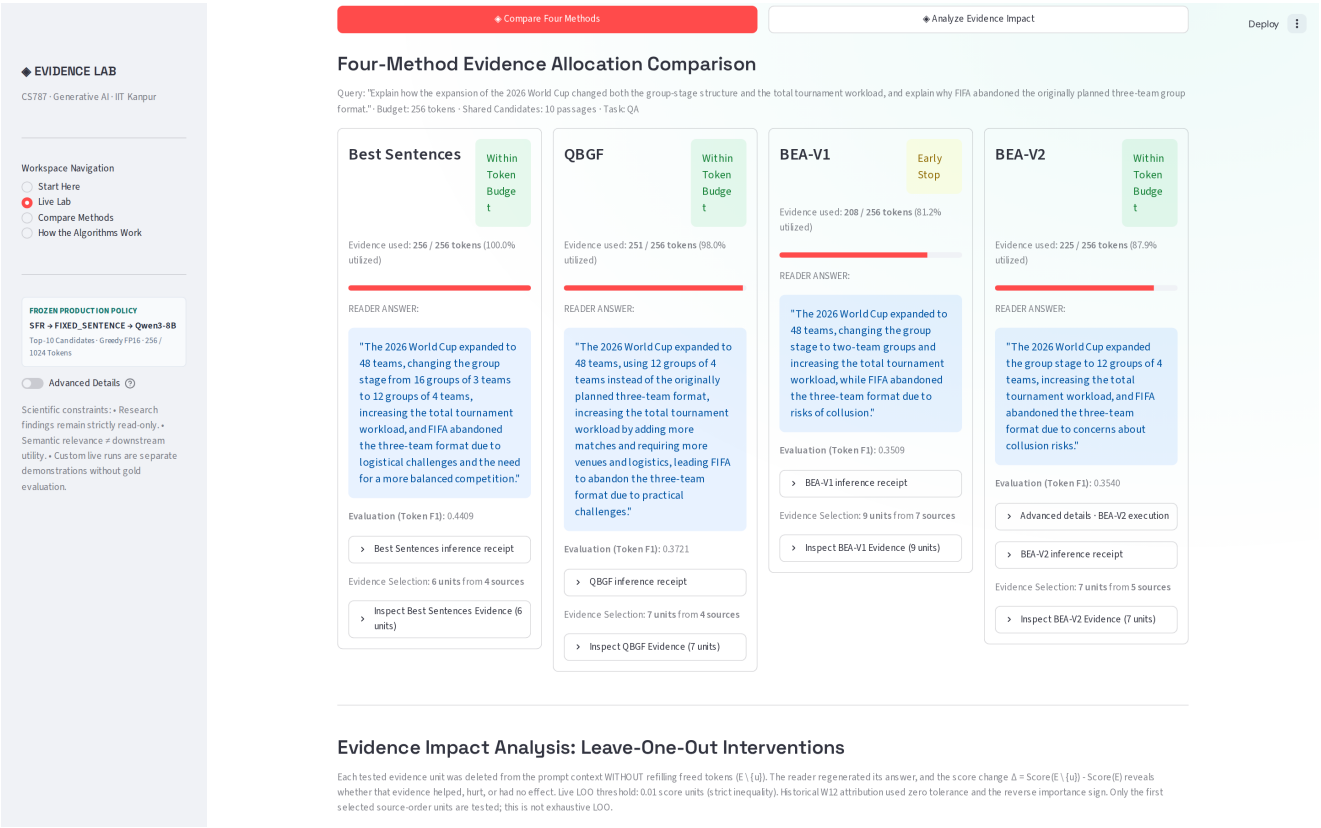


Figure 15: Live four-method comparison. FIXED_SENTENCE (displayed as "Best Sentences"), QBGF, BEA-V1, and BEA-V2 process the same custom query and candidate pool under a 256-token input-evidence budget. Cards expose actual evidence usage, generated answers, and token F1 against the preset reference. These single-query scores do not establish a method ranking.

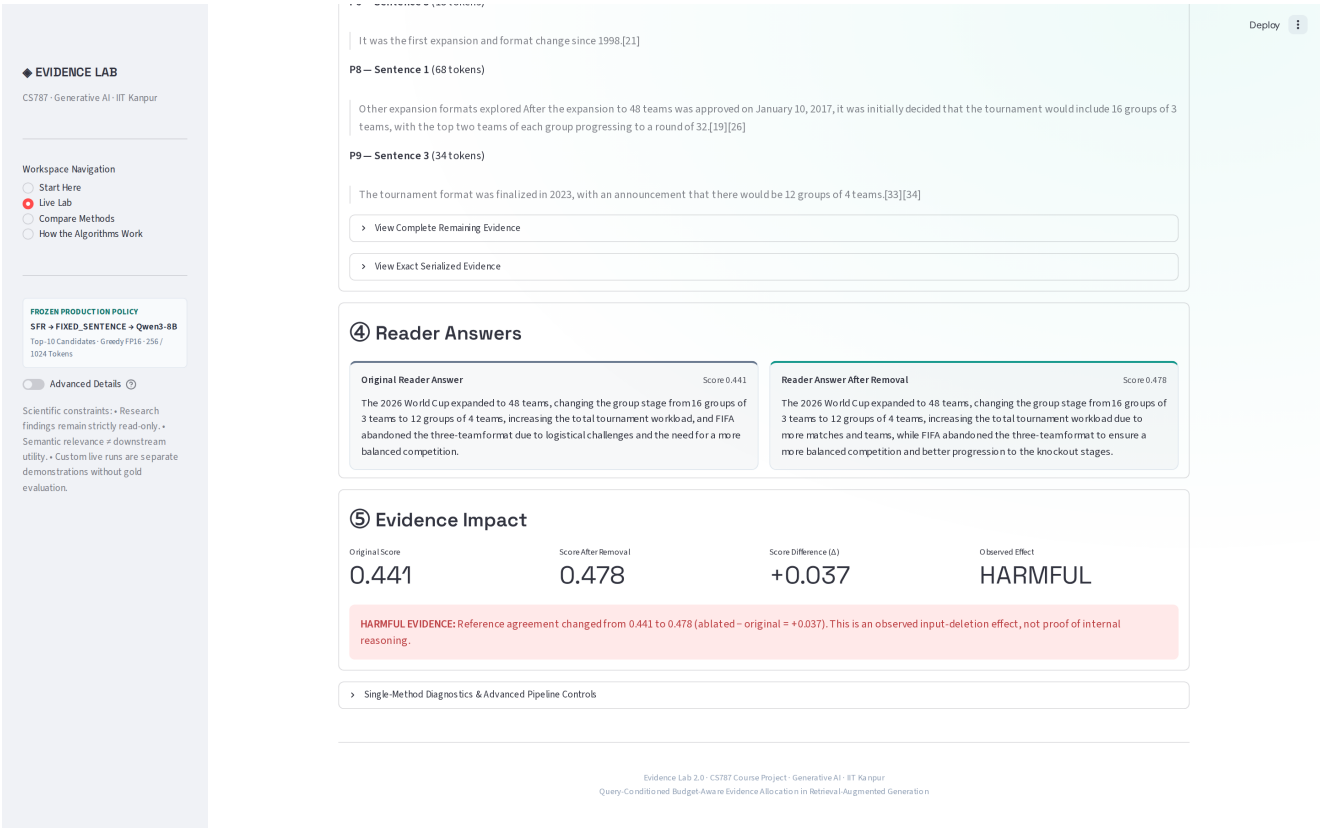


Figure 16: Live no-refill evidence deletion. For one FIXED_SENTENCE context, the reader is rerun after removing a selected unit without replacing it. Reference-agreement F1 changes from 0.441 to 0.478 (Δ = ablated − original $\approx$ +0.037). The displayed "harmful" label describes this context-specific score change, not intrinsic sentence quality, factual correctness, or proof of internal reasoning.

D Evidence-Length and Sentence Distributions

The following statistics use the same retained 450-query VAL cohort as Section 3.1. They are not full-VAL or confirmation distributions. Quantiles use linear interpolation. Passage counts are *query–passage occurrences*; a source retrieved for multiple questions contributes once per question, not once per corpus. There is no duplication across budgets.

Table 19: Detailed retrieved-pool statistics (canonical serialized evidence tokens). Mean/median/P90 and budget exceedance appear in Table 4.

Dataset	P25	P75	Maximum	Mean pool size	Queries
HotpotQA	1,121.75	1,585.5	2,327	9.95	150
QASPER	2,409.25	2,564.0	2,569	9.89	150
SciFact	2,737.00	3,926.0	6,835	10.00	150

Table 20: Individual retrieved passages: standalone source-token counts (separators excluded). A 256-token passage can cost more when serialized; standalone lengths must not be substituted for allocation costs.

Dataset	Occurrences	Mean	Median	P25	P75	P90	Max.	> 256 (%)
HotpotQA	1,492	138.2	124	85	174	227	658	5.83
QASPER	1,484	247.7	256	255	256	256	256	0.00
SciFact	1,500	332.2	303	233	396	515	1,589	66.47

Table 21: Frozen sentence units per retrieved passage. Native source sentence annotations are used where available; otherwise the frozen academic offset rules are used. These are operational sentence units, not a new manual grammatical annotation.

Dataset	Mean	Median	P90	Maximum
HotpotQA	5.13	5	8	22
QASPER	9.30	9	12	21
SciFact	9.49	9	14	43

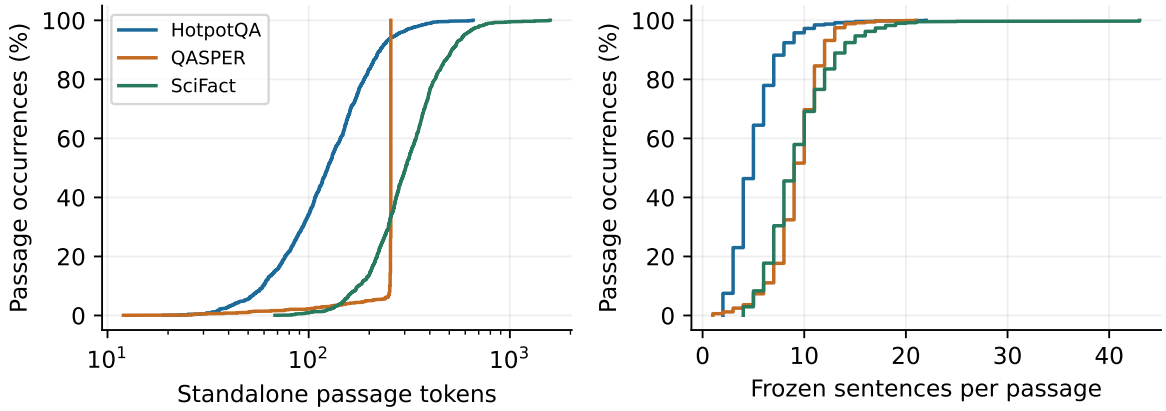


Figure 17: Passage-token and frozen sentence-count distributions on retained candidate-pool occurrences. Left: standalone passage tokens (log scale). Right: sentence units per passage.

Reproducibility and limits. The supporting distribution tables include per-query evidence lengths, passage lengths, sentence counts, and tokenizer details. Retrieved-pool statistics describe only the 450 retained candidate pools; full source-document distributions are reported separately in Appendix E. Complete retrieval pools for the entire validation cohort are unavailable, so these two descriptive populations are not conflated.

E Original Dataset Distributions and Data Lineage

This appendix characterizes complete available official inputs, not the 450 retained retrieval pools. Token counts are from the pinned Qwen tokenizer, with no maximum-length truncation and no reader special tokens. HotpotQA source paragraphs exclude titles for standalone lengths; complete question contexts include titles and double-newline boundaries. QASPER full supplied text includes title, abstract and section text; a missing section heading is empty, and one TRAIN paper with no body sections remains included. SciFact uses abstract bodies, not full papers. Additional descriptive statistics include P10/P25/P75/P95/P99 source-length quantiles, query and answer lengths, paragraph and sentence counts, questions per source, and cited-context distributions.

Table 22: Complete available split-wise original source lengths; repeated sources are counted once within each split. SciFact cited subsets are descriptive associations, not the retrieval search corpus.

Dataset	Split/unit	*N*	Total tokens	Median	P75	P95	Max
HotpotQA	dev_distractor	66,635	8,532,650	115	161	258.0	1,985
HotpotQA	test_fullwiki	66,601	8,644,141	116	163	260.0	1,586
HotpotQA	train	483,696	56,026,248	104	146	231.0	1,985
QASPER	test	416	2,053,538	4,649	5,881	8,681.2	30,009
QASPER	train	888	4,897,887	5,164	6,510	10,490.6	36,505
QASPER	validation	281	1,420,157	4,813	5,974	8,731.0	22,260
SciFact	dev	283	97,496	271	435	722.0	1,409
SciFact	shared_corpus	5,183	1,601,293	282	378	559.0	1,925
SciFact	train	565	198,861	275	443	760.2	1,409

Populations, associations and missing labels. HotpotQA unique paragraph counts are 483,696 / 66,635 / 66,601 for TRAIN / distractor DEV / fullwiki TEST; their union is 535,218 supplied paragraph identities, not all of Wikipedia. QASPER has 888/281/416 papers; sentence chunks are downstream units, not additional papers. SciFact corpus size is 5,183; the 565 TRAIN-cited and 283 DEV-cited abstracts are subsets of that shared corpus. Its full TRAIN/DEV claims form 489/247 transitive citation components within those splits (components are built jointly to prevent cross-role links), not 489/247 unique papers. Test cited IDs and annotations are unavailable. SUPPORT/REFUTE/NEI counts are 332 / 173 / 304 TRAIN and 124 / 64 / 112 DEV. Original query IDs are disjoint across splits; repeated normalized question text is documented separately and is not automatically source leakage.

Project lineage. Historical SARA internal development (231 questions/86 papers), FIT (1,746) and INTERNAL-VAL (342) derive from official QASPER TRAIN after legacy answer filtering; official QASPER validation is a different 1,005-question population. Historical answerable QASPER TEST (1,309 queries) has already been consumed and is permanently closed to new confirmation. The final TRAIN-eligible membership contains 90,447 HotpotQA, 2,178 QASPER (790 papers) and 531 SciFact examples; this is eligibility metadata, not proof of LoRA training (the full adaptation stages were skipped). Main VAL contains 7,155/1,005/177 queries, with 157 SciFact components covering 187 cited document IDs. Reserved confirmation is 200 HotpotQA plus 50 SciFact (43 groups), with no fresh QASPER cohort. Exploratory attribution/BEA pilot queries overlap VAL by design (50 per dataset); group-aware cross-fitting does not transform them into untouched data.

Dataset sources and provenance. HotpotQA counts and contexts use the official distractor training/development inputs and the separately identified public fullwiki test inputs. QASPER composition and full-paper lengths use the released train, validation, and test splits; SciFact uses its released claims and shared abstract corpus. Source-document statistics and retrieved-pool statistics are reported separately. These descriptive measurements do not constitute new held-out reader evaluations.

F Helpfulness Classifier: Features and Diagnostics

The following ordered schema is shared by training and inference. Features are finite floats; missing or nonfinite values fail rather than being imputed. SFR similarities use the frozen embedding space. StandardScaler is fitted only on the other four folds; no class weights or outcome-feature inputs are used. Each observation is a historical selected-sentence intervention, so class prevalence is conditional on the historical selector/context.

Table 23: Exact 12-feature classifier order (repository names abbreviated only for layout).

No.	Feature	Definition
1	Query–sentence cosine	SFR cosine of candidate sentence and query.
2	Passage–query cosine	Parent passage relevance.
3	Retrieval rank	One-based parent rank in the fixed retrieved pool.
4	Within-parent relevance rank	One-based sentence rank by query cosine, source-order tie break.
5	Normalized source position	Zero-based sentence index divided by $\max(1, n_{parent} - 1)$.
6	Sentence token cost	Standalone sentence Qwen token count.
7	Passage token cost	Standalone parent Qwen token count.
8	Parent sentence count	Number of frozen sentence units in parent.
9	Budget label	256 or 1024; the observation retains its intervention budget.
10	Candidate-pool width	Actual number of retrieved passages.
11	Mean pool similarity	Mean pairwise cosine to other candidate sentences (zero if none).
12	Maximum pool similarity	Maximum pairwise cosine to other candidates (zero if none).

Table 24: OOF diagnostic hard classifications at probability 0.5. Confusion order is TN/FP/FN/TP. This cutoff is descriptive; allocation uses the continuous probability.

Dataset	Precision	Recall	F1	Balanced accuracy	Confusion
HotpotQA	.0000	.0000	.0000	.5000	2753 / 0 / 188 / 0
QASPER	.6818	.0452	.0847	.5215	3204 / 7 / 317 / 15
SciFact	.0000	.0000	.0000	.5000	2018 / 0 / 35 / 0

The 8,537 observations contain 555 helpful, 552 harmful and 7,430 neutral labels. Unique sentence IDs number 1,926 for HotpotQA, 2,164 for QASPER and 1,367 for SciFact; counts repeat across historical method/budget contexts. All 15 saved models have disjoint training and held-out groups, and their source checksums match the recorded manifest. However, these predictions reuse development queries and historical deletion labels rather than testing addition utility on fresh contexts. No independent classifier-confirmation experiment or uncertainty interval over retraining was obtained. Modest AP and near-zero recall at the diagnostic cutoff prohibit interpreting P_{help} as a reliable probability of reader correctness. The displayed metrics summarize the grouped out-of-fold predictions; they are not estimates from an independent final test cohort.

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